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n8n Models

Last amended: 21st July, 2026
My folder: D:\OneDrive\Documents\n8n

A. N8n re-installation

Uninstall n8n:

- i) Remove docker container (*docker rm <cid>*)
- ii) Remove data volume (*docker volume rm n8n_data*)
- iii) Remove docker image (*docker rmi <imageId>*)

Reinstall n8n:

- i) Create a data volume:
`docker volume create n8n_data`
- ii) Run the n8n container: Execute the following (one-line) command to download and start n8n:
`docker run -it --rm --name n8n -p 5678:5678 -v n8n_data:/home/node/.n8n docker.n8n.io/n8nio/n8n`

Update n8n:

Issue command:
`./update_n8n.sh`

B. Word Embedding Visual Inspector: [Link here](#)

C. Keyboard shortcuts

Move all nodes together	Press ctrl key (or Space key) then use mouse to drag all the nodes
Move some nodes together	Press ctrl+A on each node and then move such nodes together
Save the workflow	Ctrl+S
Restart n8n	docker start n8n
Zoom in	= (equals) or + (plus) key
Zoom out	- (minus) or _ (underscore) key
Zoom to fit	1
Reset Zoom	0
Paste nodes	Ctrl+v
Add sticky node	Shift + S
Quick access to action menu	Ctrl+K

Sticky notes

Writing in Markdown

Sticky Notes support Markdown formatting. This section describes some common options.



Embed a YouTube video

To display a YouTube video in a note, use the `@[youtube](<video-id>)` directive with the video's ID. For this to work, the video's creator must allow embedding.

For example:

```
@[youtube](ZCuL2e4zC_4)
```

To embed your own video, copy the above syntax, replacing `ZCuL2e4zC_4` with your video ID. The YouTube video ID is the string that follows `vs=` in the YouTube URL.

Ollama running gguf models

Directly run a gguf model available in HuggingFace repo. See [this](#) article.

```
ollama run hf.co/<completeModelName>
```

For example, to download and run [this](#) gguf model from HuggingFace, issue command:

```
ollama run hf.co/gpustack/stable-diffusion-v1-4-GGUF
```

Ollama cloud models

To access cloud models without ollama api-key, on the terminal execute `'ollama signin'`. Then reach out the web-page and sign into it. Next `Connect` your device. Once our laptop is connected, in the terminal execute `'ollama run minimax-m3:cloud'`. Then say `/bye` to get out and from now onwards, this model will remain available as a local model. Other small models are: `gpt-oss:20b-cloud` ; `gemma4:cloud`; `nemotron-3-nano:30b:cloud`.

D. Basic LLM Chain

We use ollama to answer any questions

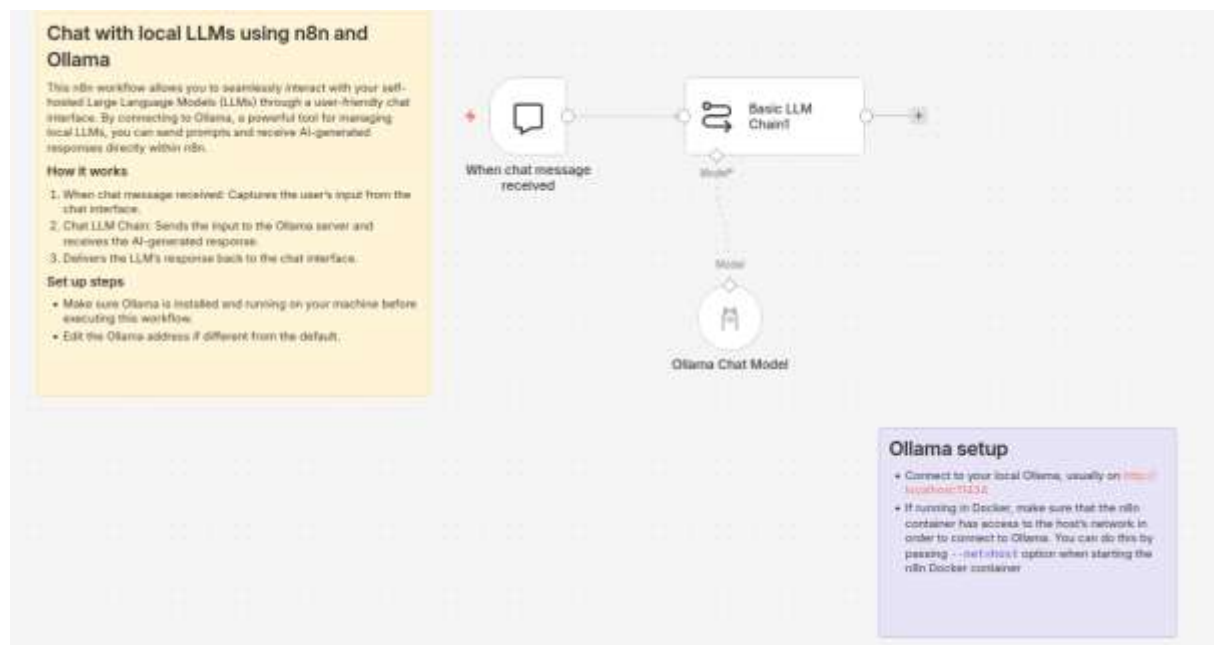


Figure 1: A basic LLM chain that simply answers questions from ollama. Use llama3.2

E. Ollama custom model: Modelfile:

Creating custom ollama models using *Modelfile* are a pain. In most cases error comes. One way to avoid the error is to work from within docker. Also, Ollama *chat model* does not allow *temperature* to be set beyond 1.0. To do that, one can create ollama custom models with preset parameter values, as:

- a. Enter your container's terminal and open bash:

```
docker exec -it ollama /bin/bash  
(This also works → docker exec -it ollama bash )
```

- b. Create the Modelfile text directly on the container's hard drive:

```
echo -e "FROM llama3.2\nPARAMETER temperature 2.0" > Modelfile
```

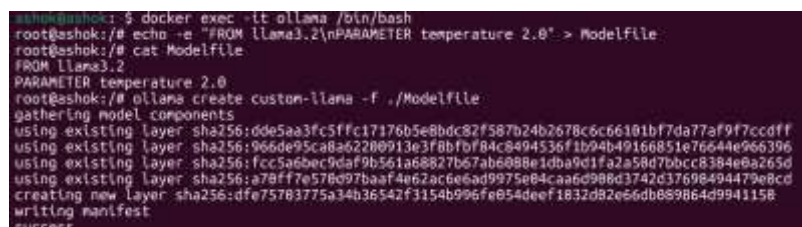
- c. Create custom model now:

```
ollama create custom-llama -f ./Modelfile
```

- d. Modelfile is:

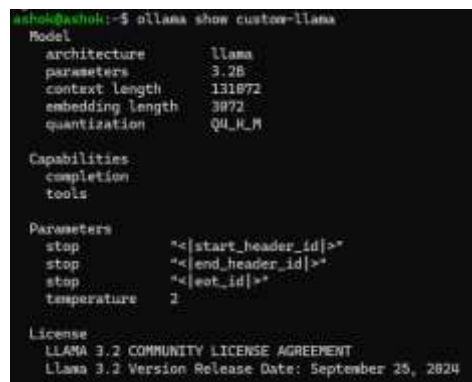
```
FROM llama3.2  
PARAMETER temperature 2.0
```

This works and custom model gets created (see figure below).



```
ashok@ashok:~$ docker exec -it ollama /bin/bash  
root@ashok:/# echo -e "FROM llama3.2\nPARAMETER temperature 2.0" > Modelfile  
root@ashok:/# cat Modelfile  
FROM llama3.2  
PARAMETER temperature 2.0  
root@ashok:/# ollama create custom-llama -f ./Modelfile  
gathering model components  
using existing layer sha256:dde5aa3fc5ffc17176b5e8bdc82f587b24b2678c6c66101bf7da77af9f7ccdff  
using existing layer sha256:966de95ca8a62200913e3f8bf84c8494536f1b94b49166851e76644e966396  
using existing layer sha256:fcc5a6bec9daf9b561a68827b67ab6088e1dba9d1fa2a58d7bbcc8384e0a265d  
using existing layer sha256:a78ff7e578d97baaf4e62ac6e6ad9975e84ca6d988d3742d37698494479e8cd  
creating new layer sha256:dfe75783775a34b36542f3154b996fe054deef1832d82e66db889864d9941158  
writing manifest  
success
```

Figure 2: Ollama custom model creation from within docker



```
ashok@ashok:~$ ollama show custom-llama  
Model  
architecture      llama  
parameters        3.2B  
context length    131072  
embedding length  3872  
quantization      Q4_K_M  
  
Capabilities  
completion  
tools  
  
Parameters  
stop              "<[start_header_id]>"  
stop              "<[end_header_id]>"  
stop              "<[eot_id]>"  
temperature       2  
  
License  
LLAMA 3.2 COMMUNITY LICENSE AGREEMENT  
Llama 3.2 Version Release Date: September 25, 2024
```

Figure 3: To see details of a model, use 'ollama show' command

F. temperature and top_p experiment

Folder: C:\Users\ashok\OneDrive\Documents\n8n\14.temperature\

Experiment design is simple. For *ollama chat model*, vary parameters and check response to user prompt written below. Creativity increases when both *temperature* and *top_p* are high. When *temperature* (0.1) and *top_p* are low (0.1), output is repetitive. (In Ollama, the *temperature* setting controls the **randomness and creativity** of a model's responses by scaling the probability of potential next words or effectively dividing current word probabilities by temperature-value.)

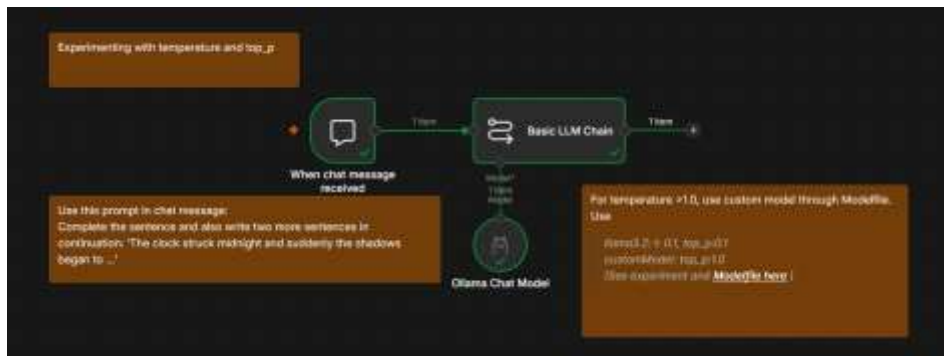


Figure 4: Ollama chat model does not permit temperature to exceed 1.0. Therefore, create and use customModel with pre-fixed temperature of 2.0.

User Prompt:

Complete the sentence and also write two more sentences in continuation: 'The clock struck midnight and suddenly the shadows began to ...'

G. Data transformation with JavaScript

The following Workflow demonstrates, transforming fields of a fake data store using Edit Fields node. We use simple JavaScript for transformation. In n8n anything within two curly braces ({{ }}) is a JavaScript expression. Some JavaScript expressions are:

```
$json.id
Number($json.id)
$json.name + " in " + $json.country
```

Knowledge of JavaScript is helpful.

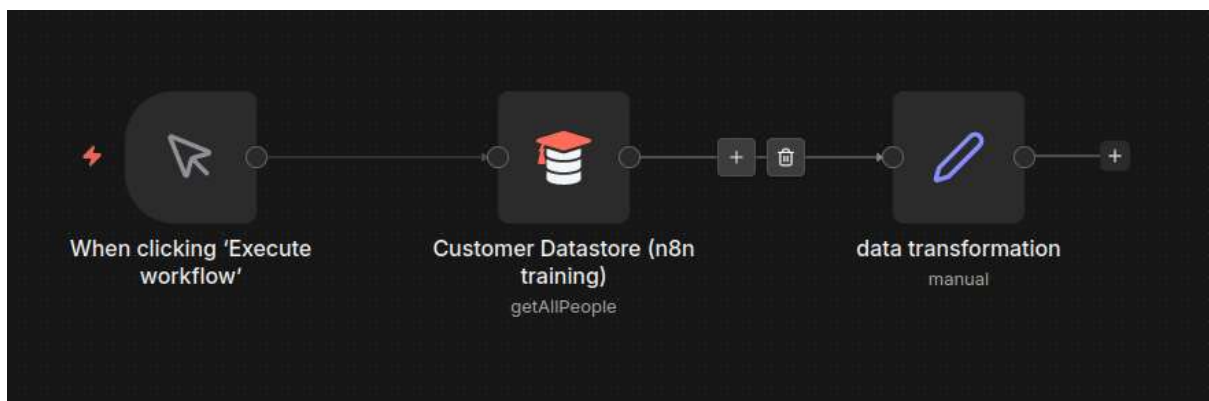


Figure 5: Simple workflow

ParametersSettings

Execute step

Mode

Manual Mapping

Fields to Set

customer_id

Number

= {{ Number(\$json.id)/100 }}

customer_name

T String

= {{ \$json.name + " in " + \$json.country }}

customer_email

T String

= {{ \$json.email }}

customer_country

T String

= {{ \$json.country }}

Drag input fields here or Add Field

☐ Include Other Input Fields

Figure 6: Data transformation using JavaScript.

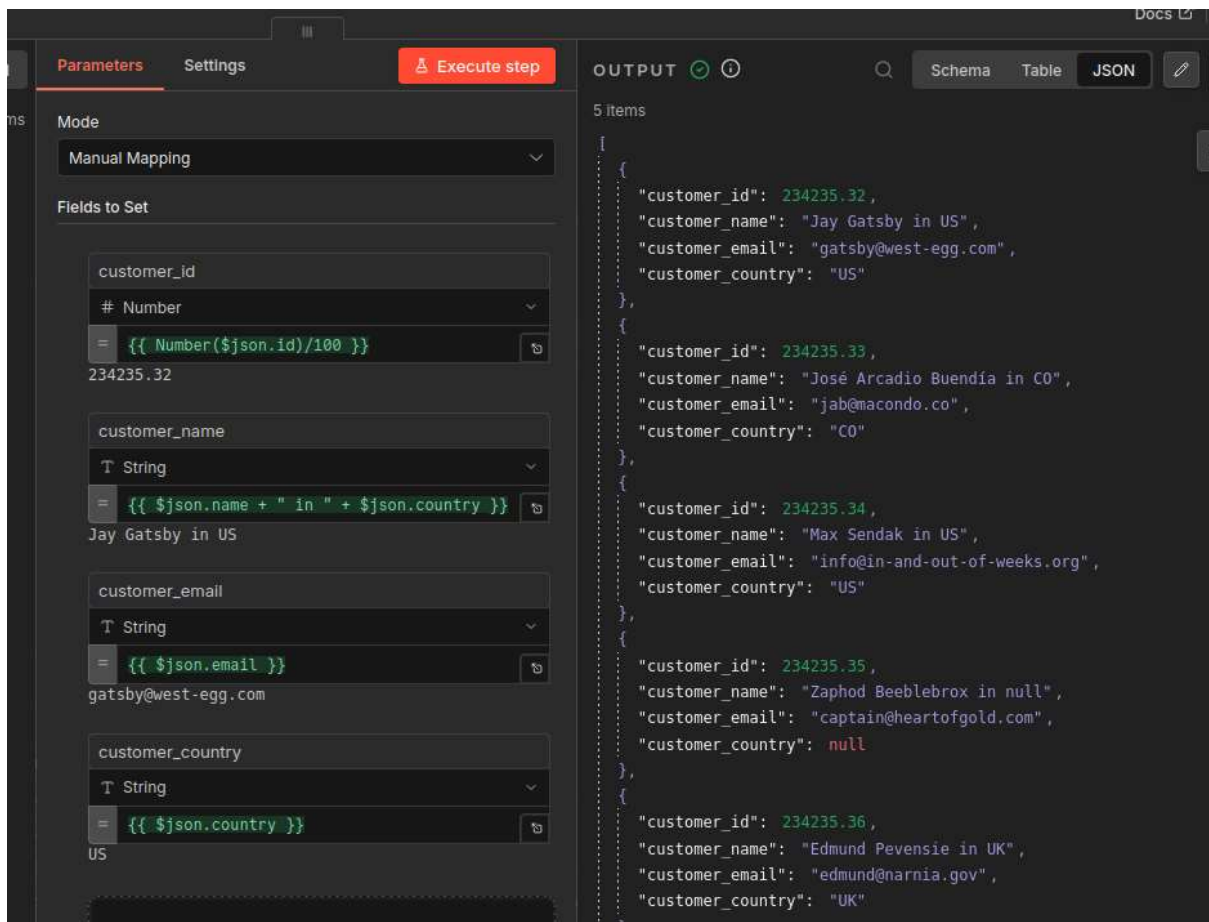


Figure 7: Right panel shows the transformed data

H. AI Agent—I

D:\OneDrive\Documents\n8n\0.aiAgent*.json

Refer [this video](#)

Our workflow:

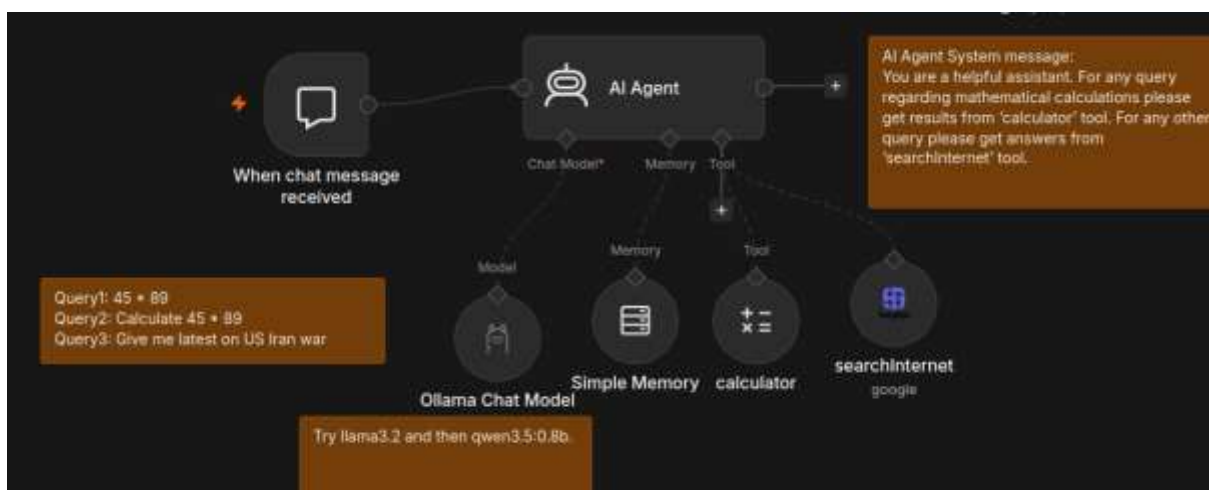


Figure 8: Use qwen3.5:0.8b (very small model). This is [SerpAPI](#)

Configuring AI Agent:

You also have to write a system prompt:

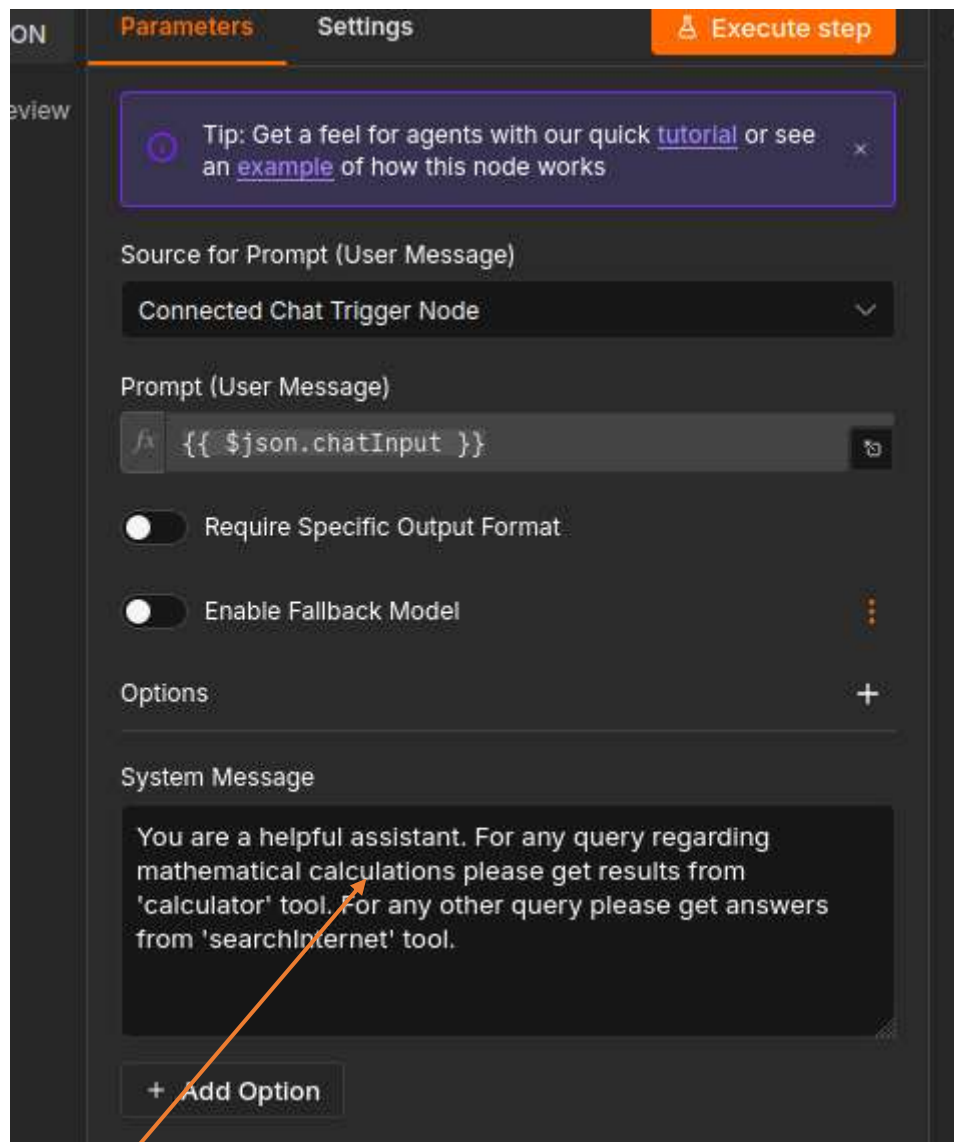


Figure 9: Write system prompt also

The configuration for Serp API is as below.

Credential

SerpApi account

Tool Description

Set Automatically

Resource

Search

Operation

Google Search

Search Query (q)

`{{ $json.chatInput }}`

Location (location)

Figure 10: Serp API configuration. Note the search Query field

System Message Template:

A recommended *System Message* template is below. See [this link](#) also.

#role

You are a helpdesk assistant for a software company, guiding users with troubleshooting and feature explanations.

#behavior

Be professional yet approachable. Do not guess answers; ask clarifying questions when needed.

#capabilities and restrictions

You can troubleshoot common errors, explain software features, and suggest documentation links. You cannot access private user data, or provide legal or financial advice.

#output

Provide direct answers with clear steps if applicable. Use bullet points for multi-step instructions. Include links to documentation where relevant.

Figure 11: Broad System Message format

Making chat public on a URL

See this diagram again and press **Publish** button. For every **change**, you have to **Publish/Save again** and give a new version number.

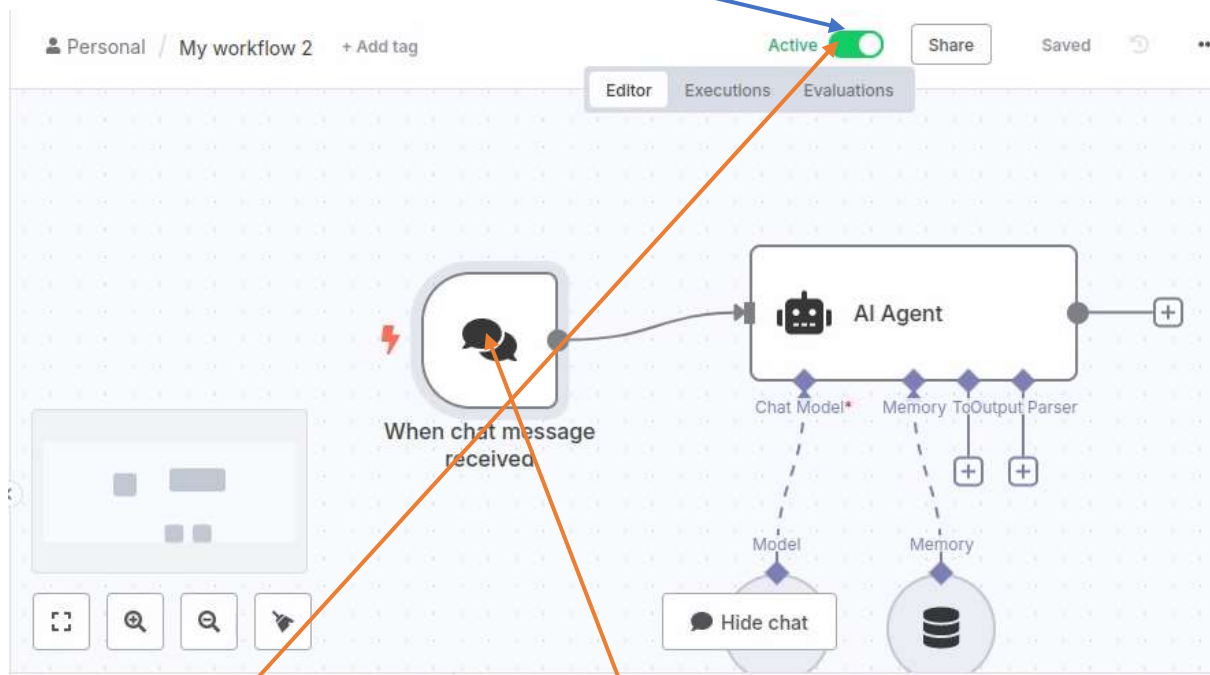


Figure 12: Make workflow **Active**. Then, double click to open the Chat message trigger and proceed to next diagram.

Double Click on the first node (*When chat message received*) to open it:

When chat messa... Test chat

Parameters Settings Docs

Chat URL

http://localhost:5678/webhook/cb6159d4-8fd4-455b-bf04-438a720c33da/chat

Make Chat Publicly Available

☒

Mode

Hosted Chat

Chat will be live at the URL above once you activate this workflow. Live executions will show up in the 'executions' tab

Authentication

None

Initial Message(s)

Hi there! 🤖
My name is Nathan. How can I assist you today?

Figure 13: Click on **Make chat public** and then get the URL

And here is the web-page with the copied URL:

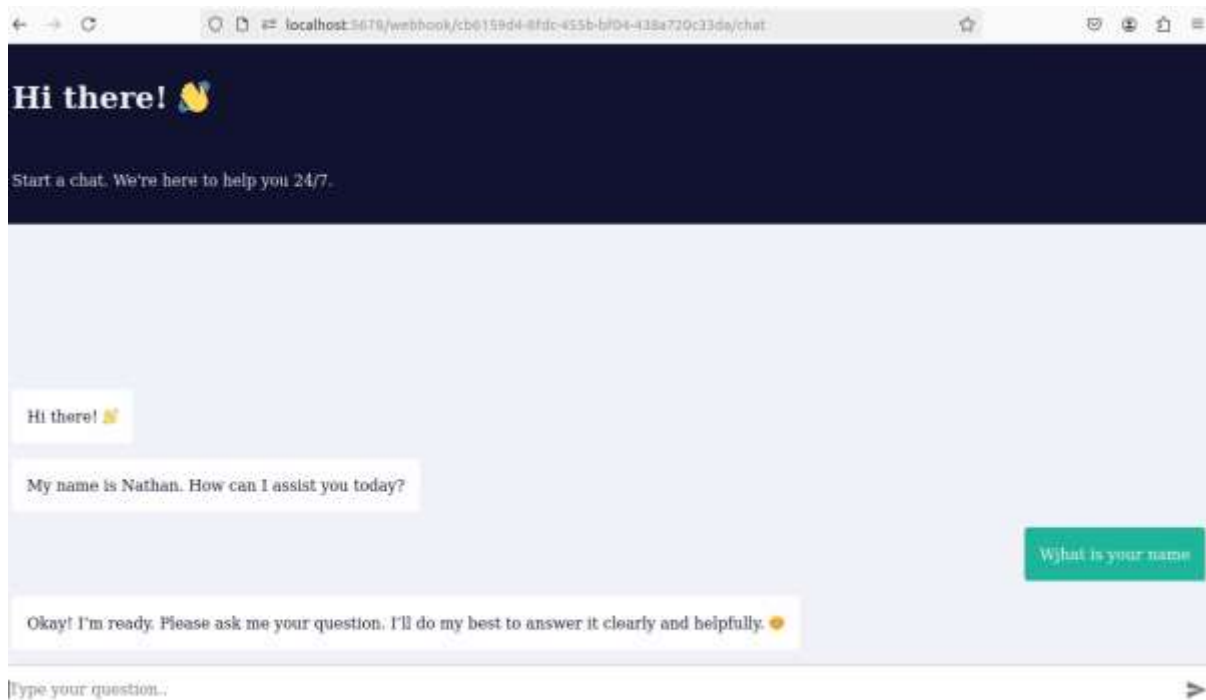


Figure 14: Chat message that triggers n8n workflow

I. AI Agent—II

(File: OneDrive\Documents\n8n\0.aiAgent\AI-Agent.json)

This agent *does not* use Chat node neither also gets its answers from Internet. Appropriate for a class-test WITHOUT Internet. When this workflow is Published, in Production mode, you can see the form but not the answer. So connect a DataTable or Postgresql node.



Figure 15: This AI Agent DOES NOT have a chat node. Neither also it answers questions from Internet. It answers questions from an LLM node used as a tool.



Figure 16: Insert a postgresql table at the end to receive data

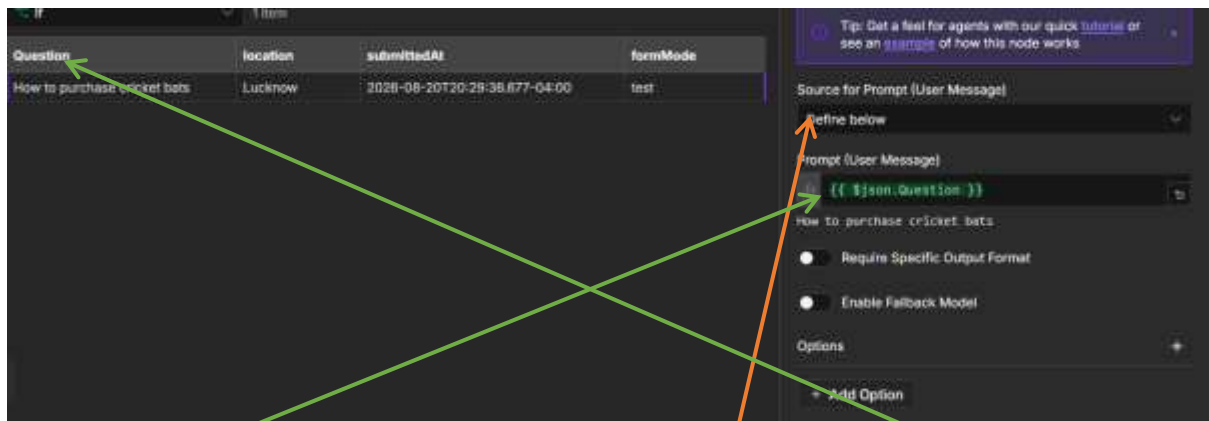


Figure 17: Prompt: AI Agent configuration (this **MUST** be selected as 'Define below'). And drag Question from the left panel to Prompt. You can, of course, write appropriate System messages.

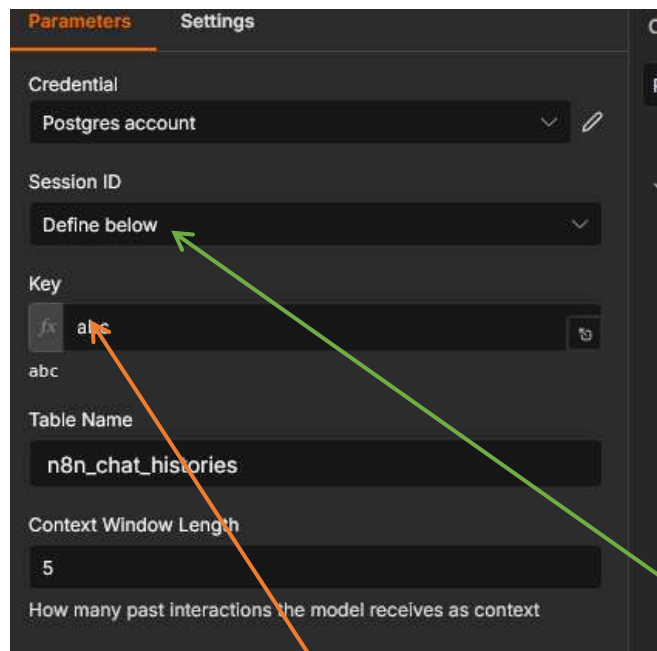


Figure 18: Session ID: Postgres memory needs a Session ID key. When input is NOT chat box, then, select Session ID as Define below and give some arbitrary id key value.

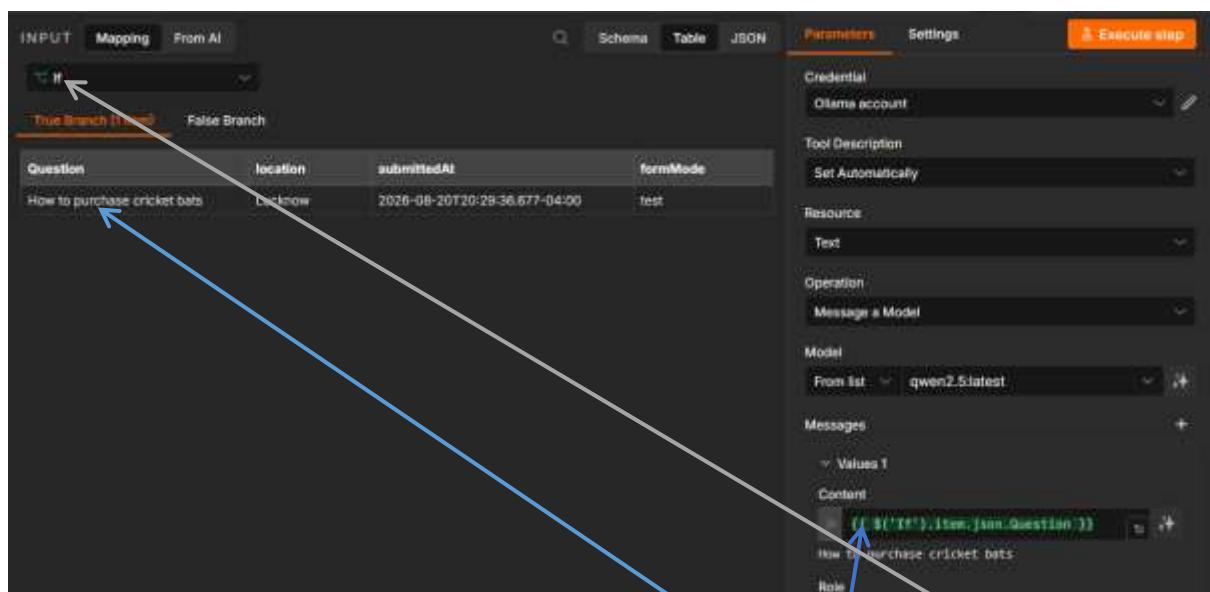


Figure 19: Ollama message model: Drag Question to Messages as 'values1'. Note the 'if' node selected from drop-down.

Blog writer in the chain:

You can also add a blog-writer in this chain as below. Blog writer is a 'Basic LLM Chain' node. Elsewhere, I have used AI Agent as Blog Writer Agent. See Blog writer agent.

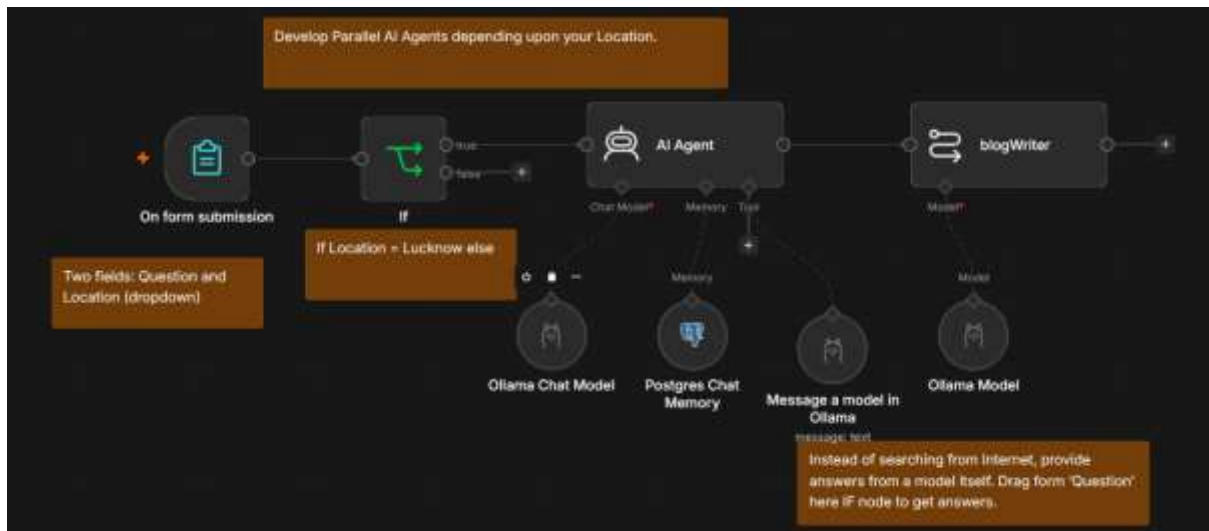


Figure 20: Instead of PostgreSQL as end-node, you can also have a blog-writer.

J. AI Agent with sub-agents

This Ai Agent uses multiple subagents as tools. But needs to be refined. All messages descriptions are given in the file: 'agentic systems with subsystems.txt'. Also .json file is there. See folder: 'OneDrive\Documents\n8n\0.aiAgent'.



Figure 21: AI agent with AI sub-agents used as tools . More work need to be done here. MAYBE better models.

K. SMTP node-I

Refer [here](#) and [here](#)

Follow these steps to send an email to your Gmail account:

- In *Gmail*, click on your Profile picture
- Click *Manage Your Google Account*→*Security and Sign in*
- Click *2-Step Verification*. Enable it. And then down below, create *App Password*

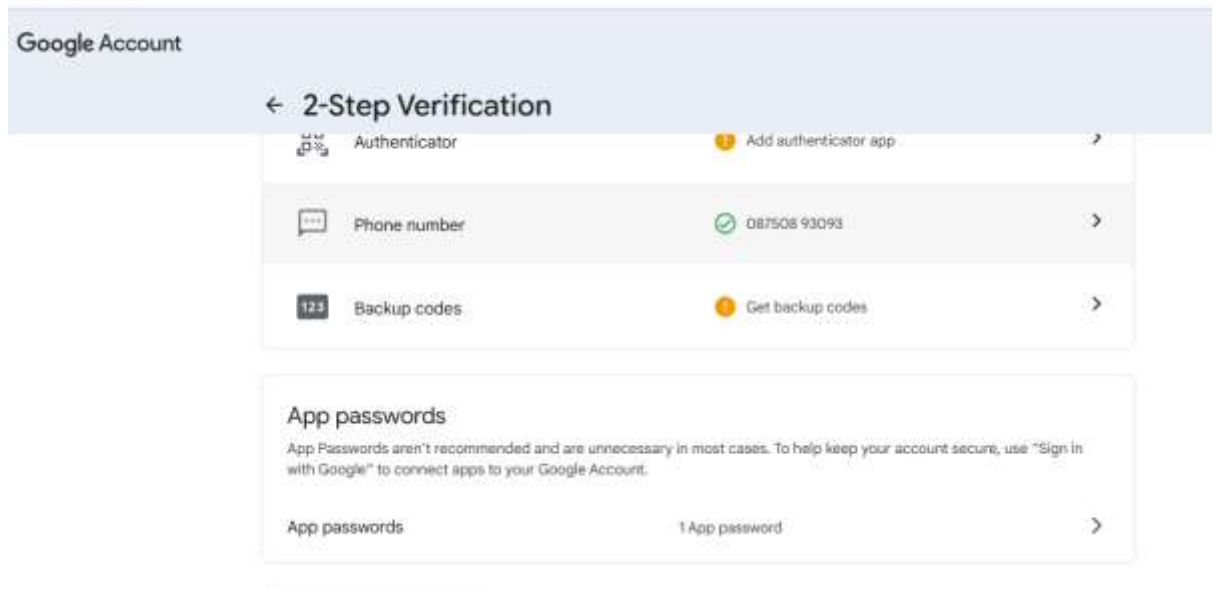


Figure 22: Creating App Password.

- a. In your Gmail account, set-up two step verification. See [this link](#) as to how to set-up two-step verification account.
- b. To generate an app password:
 - a. In your Google account, go to [App passwords](#).
 - b. Enter an **App name** for your new app password, like 'n8n credential'.
 - c. Select **Create**.
 - d. Copy the generated app password. You'll use this in your n8n credential.
- c. Here is how you setup SMTP node:

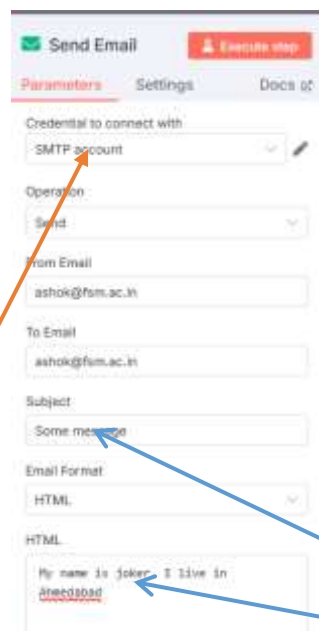


Figure 23: Set up SMTP account as below. From email should correspond to your app-password. This message will be sent

Here is how you setup SMTP account credentials.

SMTP account

Connection

Sharing

Details

Connection tested successfully

Need help filling out these fields? [Open docs](#)

User [Test](#) [Remember](#)

Password

Host

Port

SSL/TLS ☒

Send Email

Figure 24: Setup SMTP credentials: User is your email address. Password is App-password. Host is always: **smtp.gmail.com**. Port is always: **465**. Client Host name is NULL.

Send Email

Parameters

SMTP account

From Email

To Email

Subject

Email message

Email Format

HTML

My name is jason. I love to Overkill

Execute

OUTPUT

accepted	rejected	area	errorCode	message
1	0	SMTP	0	My name is jason. I love to Overkill

Figure 25: On click Execute step an email will be sent to the send account. This email does not carry calculator output. It has fixed message. See below

Send calculator output:

Send email Execute step

Parameters Settings Docs

Credential to connect with
SMTP account

Operation
Send

From Email
cdsht@fsm.ac.in

To Email
ashok@fsm.ac.in

Subject
easy task

Email Format
Text

Text
fx {{ \$json.output }}

The answer to 2 + 2 is 4.

Options
No properties
Add option

Figure 26: Use Text format. And drag the output to here from left panel. In ashok@fsm.ac.in check All Mail as also Inbox

A. SMTP node--II

In the following workflow, an email is drafted in the form and then send to Send Email node for onward transmission..



Figure 27: On form submission, the email is sent to desired address.

Form titled "Email draft" with the following elements:

- Field: emailAddress * (value: ashok@bnu.ac.in)
- Field: subject * (value: Test email)
- Field: body * (value: Testing if workflow works)
- Button: Submit

Footer: Form automated with **ODG** **ma**

Figure 28: The form. Its elements are: email, text and text

These are the form elements:

The image shows a 'Form Elements' panel with three form fields. Each field has a 'Field Name' input, an 'Element Type' dropdown, a 'Placeholder' input, and a 'Required Field' toggle switch.

Field Name	Element Type	Placeholder	Required Field
emailAddress	Email	xyz@gmail.com	Yes
subject	Text	Write a subject here	Yes
body			

Figure 29: Form fields

On form submission Execute step

Parameters Settings Docs 12

Form URL

Test URL: <http://localhost:8080/form-test/0000007f-c34e-49d4-bb04-a201e1c2d0a1>

Authentication: None

Form Title: Email draft

Form Description: P.S. We'll get back to you soon.

Form Elements

Field Name	Element Type	Required Field
emailAddress	Email	☒
subject	Text	☒
body	Text	☒
submittedAt	Text	☒
formMode	Text	☒

OUTPUT

```

{
  "emailAddress": "ashok@fsm.ac.in",
  "subject": "Test email",
  "body": "This is the test email",
  "submittedAt": "2025-07-15T22:48:55.654-04:00",
  "formMode": "test"
}

```

Figure 30: Form elements to be submitted to SMTP node--Pinned

Set up credentials of SMTP service as before. And fill up fields as below.

Send Email Execute step

Parameters Settings Docs 12

Credential to connect with: SMTP account

Operation: Send

From Email: fsm.ac.in

To Email:

Subject:

Email Format: Text

Test:

Options: No attachments

OUTPUT

```

{
  "emailAddress": "ashok@fsm.ac.in",
  "subject": "Test email",
  "body": "This is the test email",
  "submittedAt": "2025-07-15T22:48:55.654-04:00",
  "formMode": "test"
}

```

Figure 31: Left panel contains data submitted from **form**. Drag fields from left panel to the Central panel instead of writing json code. For example, drag **emailAddress** from left panel and drop it in **To:Email** field; similarly for **subject** and other fields. You can press <shift><ENTER> to enter a new line.

On Execute workflow, an email goes to ashok@fsm.ac.in.

B. SMTP node--III

You can make the above workflow a little more complex by introducing a switch and in the form element writing an email.

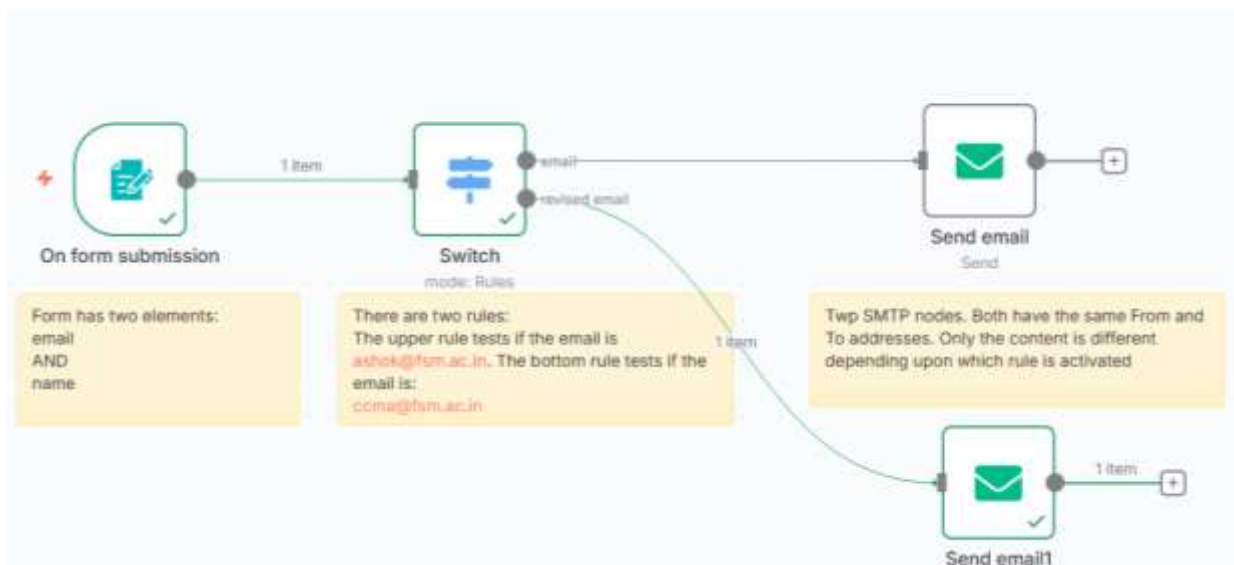


Figure 32: Configuration for switch node is as below.

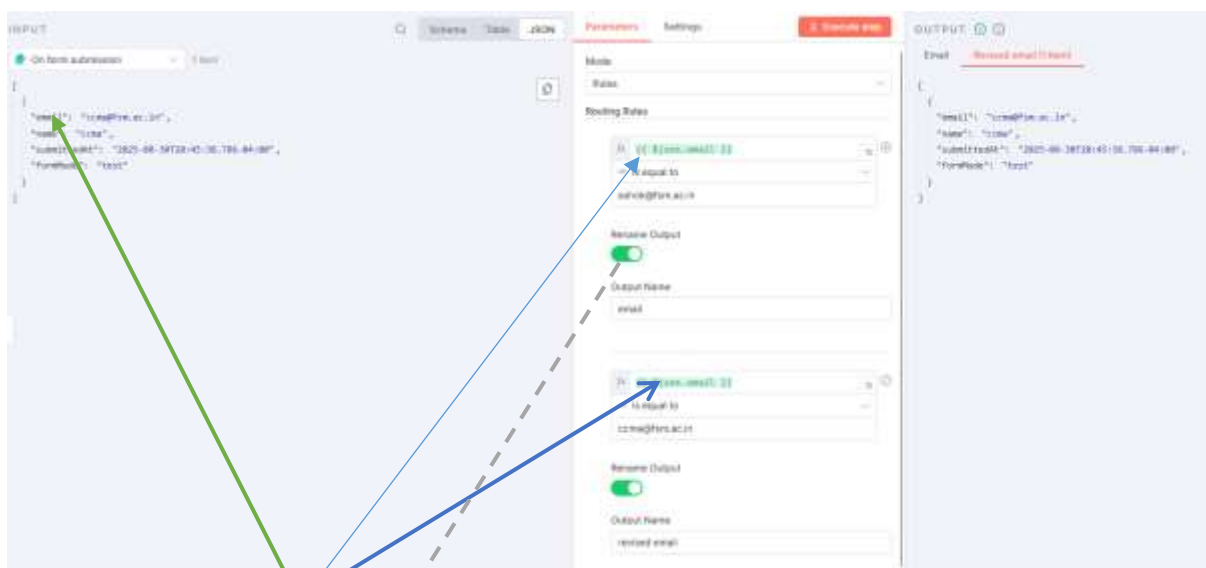
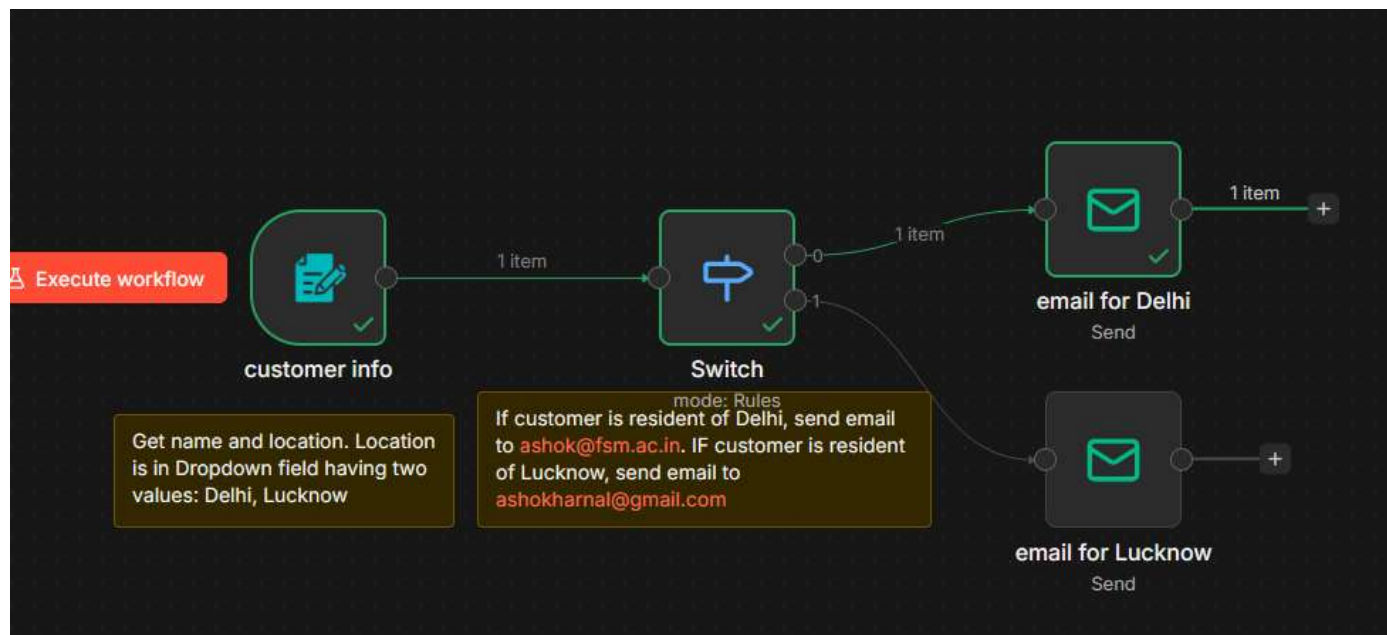


Figure 33: Input element of email is tested in the two rules. Upper rule tests for ashok@fsm.ac.in and the lower rule tests for ccma@fsm.ac.in. In both cases, **rename** your branches appropriately.



Parameters

Settings

Execute step

Mode

Rules

Routing Rules

fx

{{ \$json.email }}

AB is equal to

ashok@fsm.ac.in

Fixed Expression

Rename Output

☒

Output Name

email

fx

{{ \$json.email }}

AB is equal to

ccma@fsm.ac.in

Fixed Expression

Rename Output

☒

Output Name

revised email

Figure 34: Switch node details

Another sample rule:

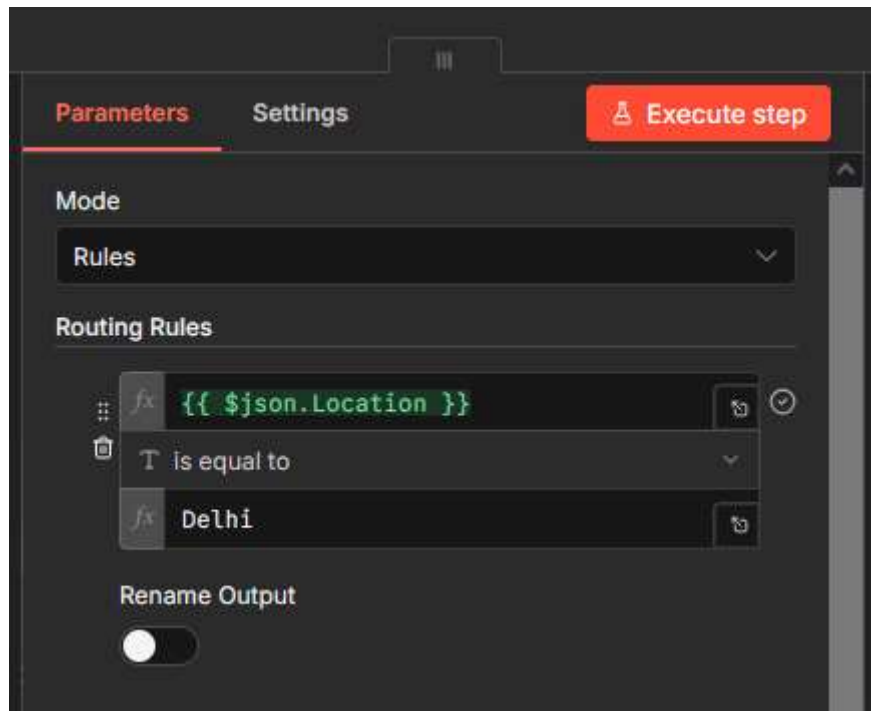


Figure 35: Note Delhi is NOT in double quotes

C. N8n memory error

n8n keeps data in memory while the workflows are running. Creating sub-workflows is a good idea as after a sub-workflow is executed, its memory is released. At times n8n breaks and gives memory error. Error message is about JavaScript heap memory being exceeded. Memory needs to be increased. See [this link](#) and [this link](#).

One can assign more memory by changing the environment variable `--max-old-space-size`. This can be done while starting docker, as:

```
docker run -d --name n8n -p 5678:5678 -e NODE_OPTIONS="--max-old-space-size=8000"
docker.n8n.io/n8nio/n8n
```

And, if n8n is directly installed, run n8n as:

```
NODE_OPTIONS="--max-old-space-size=8000" npx n8n
```

To check memory usage, issue top command. Specifically for user ashok, issue top -u ashok :

```
top - 09:33:57 up 21 min, 1 user, load average: 0.11, 0.08, 0.02
Tasks: 52 total, 1 running, 51 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.5 us, 0.1 sy, 0.0 ni, 99.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	MEM	TIME+	COMMAND
1486	ashok	20	0	38.6g	6.9g	56640	S	14.8	22.9	1:46.01	node
1605	ashok	20	0	7800	3688	3040	S	0.3	0.0	0:00.31	top
1	root	20	0	167152	11356	7836	S	0.0	0.0	0:00.81	systemd
2	root	20	0	3040	1760	1760	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3076	1820	1760	S	0.0	0.0	0:00.00	init
61	root	10	-1	47804	14080	13280	S	0.0	0.0	0:00.11	systemd-journal
90	root	20	0	21028	5888	4608	S	0.0	0.0	0:00.10	systemd-udevd
127	systemd+	20	0	26280	18240	9120	S	0.0	0.0	0:00.05	systemd-resolve
128	systemd+	20	0	89364	7040	6240	S	0.0	0.0	0:00.05	systemd-timesyn
209	root	20	0	4300	2560	2400	S	0.0	0.0	0:00.00	cron
211	message+	20	0	8588	4000	3600	S	0.0	0.0	0:00.13	dbus-daemon
223	root	20	0	30888	18400	9920	S	0.0	0.1	0:00.08	networkd-dispat
226	syslog	20	0	222404	4800	4120	S	0.0	0.0	0:00.02	rsyslogd
232	root	20	0	15336	6888	6200	S	0.0	0.0	0:00.08	systemd-logind
253	root	20	0	3240	2888	2080	S	0.0	0.0	0:00.00	agetty
257	root	20	0	3196	1920	1920	S	0.0	0.0	0:00.00	agetty
258	root	20	0	15436	9120	7520	S	0.0	0.0	0:00.00	sshd
322	root	20	0	187164	20060	13120	S	0.0	0.1	0:00.07	unattended-upgr
359	postgres	20	0	215888	20600	27040	S	0.0	0.1	0:00.05	postgres
441	postgres	20	0	215888	7048	4880	S	0.0	0.0	0:00.00	postgres
442	postgres	20	0	215888	8568	5920	S	0.0	0.0	0:00.04	postgres
443	postgres	20	0	215888	11448	8080	S	0.0	0.0	0:00.01	postgres
444	postgres	20	0	216376	10168	7200	S	0.0	0.0	0:00.00	postgres

Figure 36: top command output. RES shows total memory usage by user 'ashok' and the process is node (ie nodejs).

```

top - 09:36:41 up 24 min, 1 user, load average: 0.02, 0.05, 0.02
Tasks: 52 total, 1 running, 51 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 30935.2 total, 22917.0 free, 7765.6 used, 252.6 buff/cache
MiB Swap: 8192.0 total, 8192.0 free, 0.0 used, 22848.6 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR     S    %CPU  %MEM     TIME+ COMMAND
 1486 ashok    20   0   38.7g   0.9g   56648 S    2.8   23.0   1:48.97 node
 509  ashok    20   0   6344    512    3368 S    0.0   0.0   0:00.05 bash
 604  ashok    20   0   17096   9440   7040 S    0.0   0.0   0:00.03 systemd
 605  ashok    20   0   168892  5040   2400 S    0.0   0.0   0:00.00 (sd-pam)
 613  ashok    20   0   6236   4888   3280 S    0.0   0.0   0:00.01 bash
1473  ashok    20   0   4784    320    3040 S    0.0   0.0   0:00.00 start_nfs.sh
1474  ashok    20   0   1099032 64868  41320 S    0.0   0.2   0:00.43 npm exec nfs
1485  ashok    20   0   2896    168    1688 S    0.0   0.0   0:00.00 sh
1504  ashok    20   0   6176   5120   3368 S    0.0   0.0   0:00.01 bash
1524  ashok    20   0   7000    360    3040 S    0.0   0.0   0:00.04 top
1535  ashok    20   0   6176   5120   3368 S    0.0   0.0   0:00.01 bash
1553  ashok    20   0   5928   4096   2720 S    0.0   0.0   0:00.03 man
1561  ashok    20   0   3736    240    2000 S    0.0   0.0   0:00.00 pager
1587  ashok    20   0   6176   5120   3368 S    0.0   0.0   0:00.01 bash
1605  ashok    20   0   7888    368    3048 R    0.0   0.0   0:00.41 top

```

Figure 37: Output of command: `top -u ashok`. You can also see total memory as also free memory

D. CRUD operations on PostgreSQL

Steps:

a. Reconfigure PostgreSQL security:

- Configure `/etc/postgresql/16/main/postgresql.conf`. Here `'/16/'` is the version number. Check the line:

```
cat /etc/postgresql/16/main/postgresql.conf | grep listen
```

You will get:

```
#listen_addresses = 'localhost'
```

- Modify the `listen_addresses` parameter to allow connections from external interfaces. To allow connections from all interfaces, set it to `'*'`.

```
listen_addresses = '*'
```

- Configure `/etc/postgresql/16/main/pg_hba.conf`. For broader access (e.g., from a specific IP range), you can use a CIDR notation. In google you can raise a query: *'how to write 172.30.109.200 with netmask of 255.255.240.0 in cidr notation'*. Or see [this link](#) for calculation:

```
host all all 172.30.96.0/20 md5
```

- Restart postgresql

```
sudo systemctl restart postgresql
```

- Create a user, say kumar, and assign him necessary privileges: (See [this file](#) (at line no 170-211)for all the quick code):

```

sudo useradd -m kumar
sudo passwd kumar
sudo -u postgres psql -c 'create database kumar;'
sudo -u postgres psql -c 'create user kumar;'
sudo -u postgres psql -c 'grant all privileges on database kumar
to kumar;' -d kumar
sudo -u postgres psql -c "alter user kumar with encrypted
password 'kumar';"
sudo -u postgres psql -c " GRANT ALL ON SCHEMA public TO kumar;"
-d kumar
sudo -u postgres psql -c " CREATE EXTENSION vector;" -d kumar
# Add a table and a record

```

```
sudo -u kumar psql -c "CREATE TABLE acars ( brand VARCHAR(255),
model VARCHAR(255), year INT);" -d kumar
sudo -u kumar psql -c "INSERT INTO acars (brand, model, year)
VALUES ('Ford', 'Mustang', 1964);" -d kumar
```

- OR as: Create a table in database *kumar*, as:

```
$sudo su kumar
$psql kumar
kumar=> \c kumar
You are now connected to database "kumar" as user "kumar".
kumar=> CREATE TABLE acars ( brand VARCHAR(255), model VARCHAR(255), year INT);
kumar=> INSERT INTO acars (brand, model, year) VALUES ('Dzire', 'Maruti', 1994);
kumar=> INSERT INTO acars (brand, model, year) VALUES ('Swift', 'Maruti', 1984);
kumar=> select * from acars ;
```

postgresql quick steps:

```
./psql.sh
create user kavita with password 'kavita' createdb;
create database kavita with owner kavita ;
grant all on database kavita to kavita ;
\l
\q
sudo useradd -m kavita
sudo su kavita
psql
\c kavita
create table goat (id int primary key, name varchar(3000)) ;
\q
./reset_n8n.sh
FORM (with two text fields: id and name)==>Postgresql node(Insert)
```

- b. In n8n create the following very simple network:

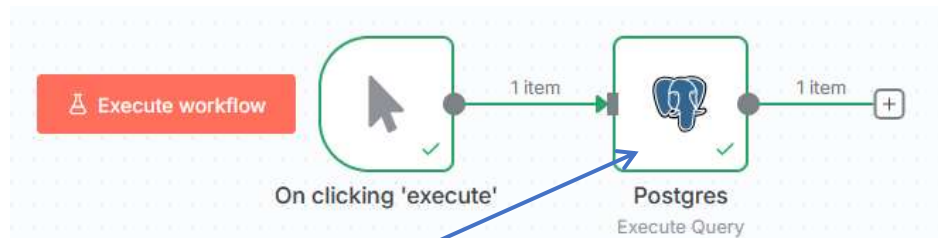


Figure 38: Double click to open it. In Query write: *select * from acars*; OR *select * from employees*;

Create PostgreSQL credentials, as:

The screenshot shows a 'Postgres account' configuration window. On the left, there are tabs for 'Connection', 'Sharing', and 'Details'. The 'Connection' tab is active, showing a green status bar at the top that says 'Connection tested successfully' with a 'Retry' button. Below this is an orange warning bar that says 'Need help filling out these fields? Open docs'. The main form contains the following fields: 'Host' with the value 'localhost', 'Database' with the value 'kumar', 'User' with the value 'kumar', 'Password' which is masked with dots, and 'Maximum Number of Connections' with the value '100'. A 'Fixed Expression' button is located to the right of the Host field.

Figure 39: User: kumar; password: kumar; database: kumar

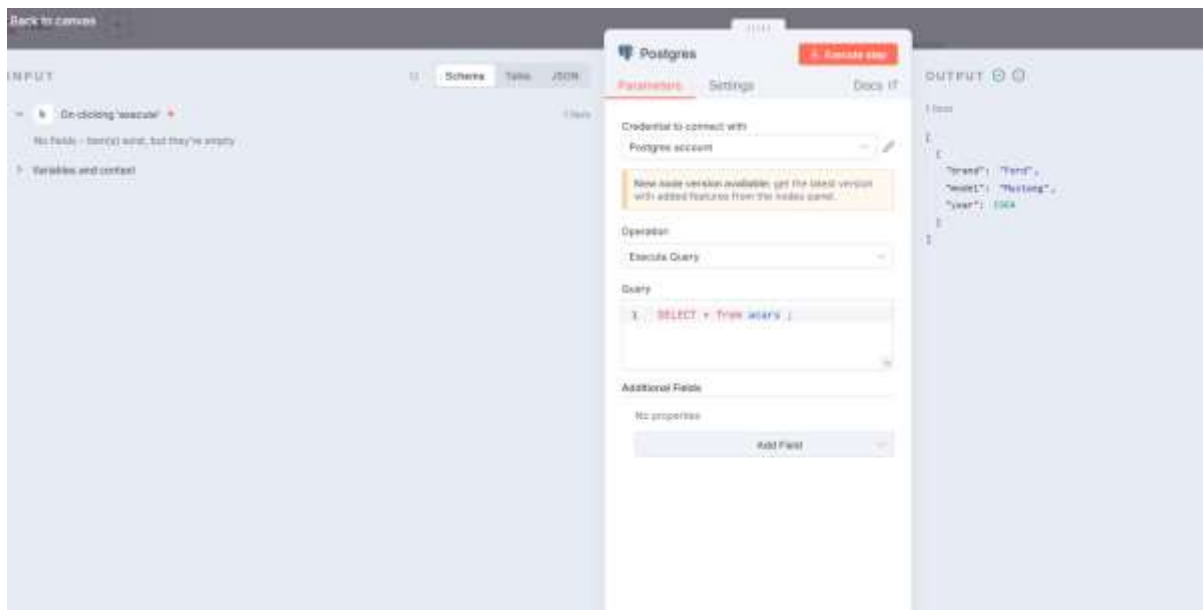


Figure 40: Query: select * from acars; The result is on the right.

Extend the above workflow further, step-by-step. Available PostgreSQL nodes are:

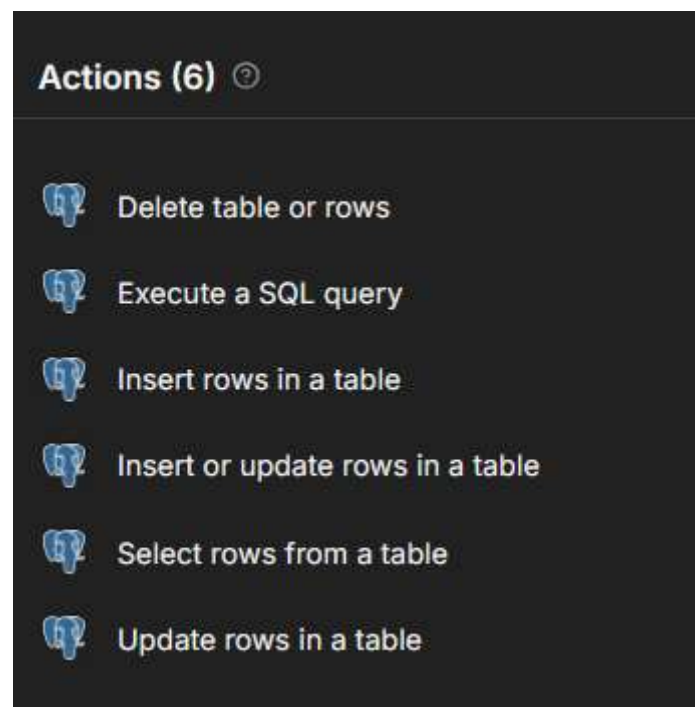


Figure 41: Available PostgreSQL nodes in n8n

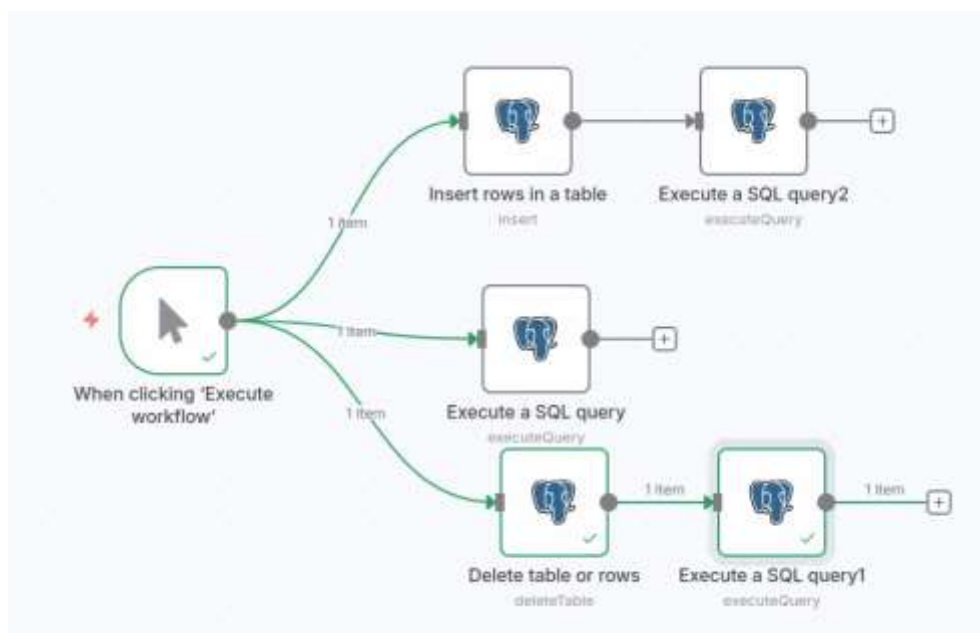


Figure 42; Workflow using different postgres nodes: Execute SQL Query, Insert rows, Delete table of rows

Delete table or rows
 Execute step

Parameters
 Settings
 Docs

Credential to connect with

gandhi

Operation

Delete

Schema

From list
 public

Table

From list
 acars

Command

Delete

Select Rows

Column
 year
 Operator
 Equal
 Value
 1971

Figure 43: Delete table rows settings

E. Postgres Data insertion with Form:

Create a table in 'ravi' database.

```
sudo sur avi
psql ravi
\c ravi
Create table ravitest (name varchar(20), email varchar(20)) ;
```

Now use form to Insert this data into postgres table. And after inserting, execute query:

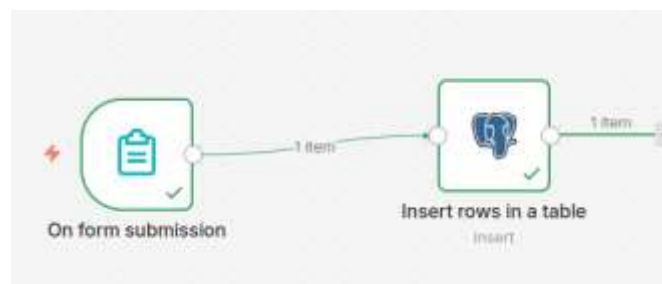


Figure 44: Form entering data into postgres

```
select * from ravitest.
```

```

ravi=> create table ravitest (name varchar(20), email varchar(20)) ;
CREATE TABLE
ravi=> select * from ravitest;
name | email
-----+-----
ashok | ashok@fsm.ac.in
(1 row)

```

Figure 45: Data inserted from form into postgres

F. SQL generation -- AI agent and postgres

Simple SQL agent

Database can be postgres, or SQLite or MySQL. In the terminal use `./psql.sh`, then connect to ravi (`\c ravi`). Then issue commands: `\l` and `\d` or `select * from spj`; *Execute Query: {{ \$fromAI('sql_statement') }}*. Model: [gwen3.5:0.8b](#) also works.

Query1: Count the number of rows in table spj ; **Query2:** List the names of tables in database ravi

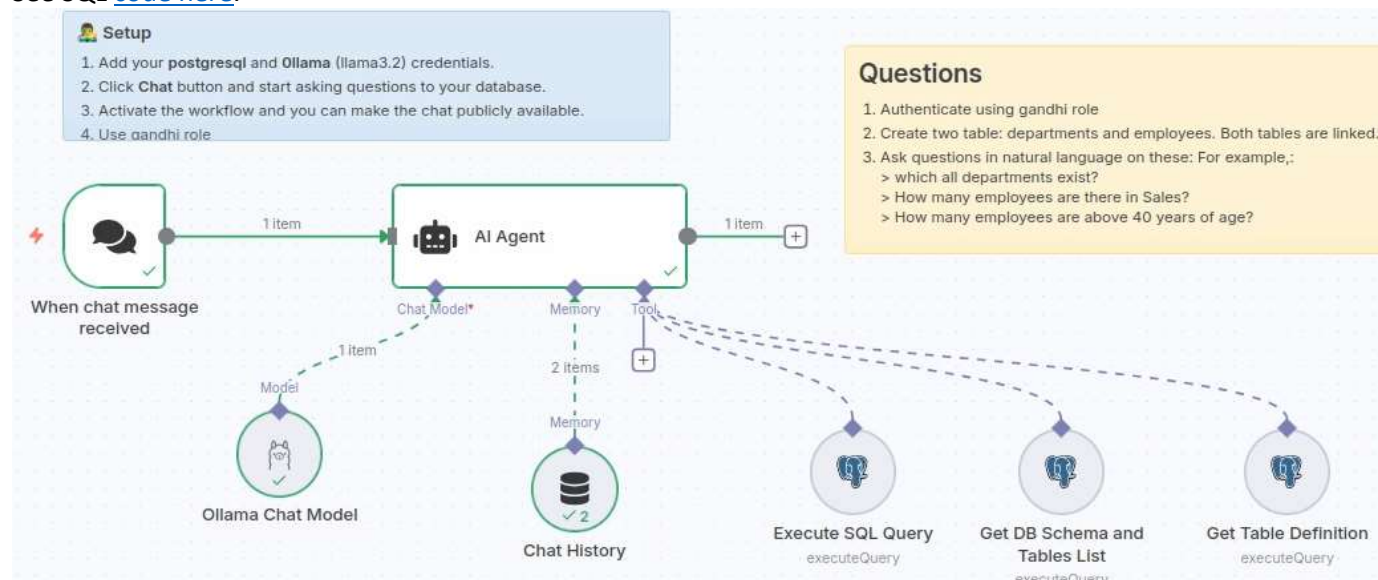


Figure 46: User is ravi, password is ravi and database is ravi. The database contains the standard C J Date's tables: s,p,j,spj and an additional table : distributors. Model: [gwen3.5:0.8b](#) also works

Complex SQL ai agent (with postgres)

<https://n8n.io/workflows/1954-ai-agent-chat/>

See SQL [code here](#).



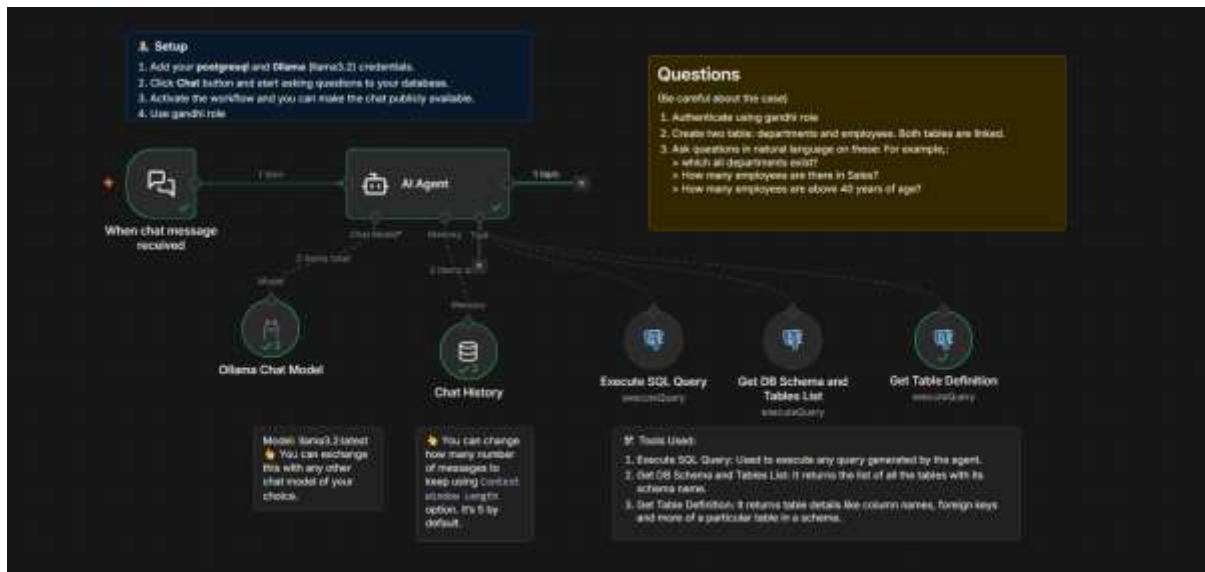


Figure 47: Another view with theme change

Replace chat history by PG Chat memory. In PG Chat memory, you can specify your own table name.

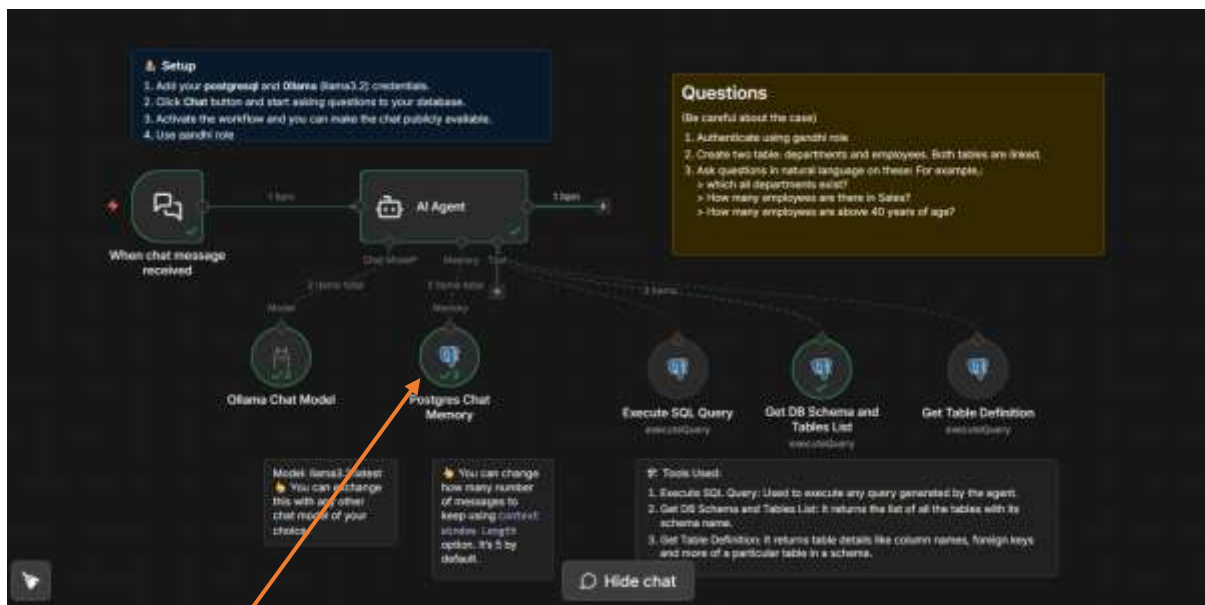


Figure 48: Replace chat history by PG Chat memory. The PostgreSQL account or credentials can be the same as in other PG nodes. In PG Chat memory, you can specify your own table name.

System prompt of AI agent:

You are DB assistant. You need to run queries in DB aligned with user requests.
Run custom SQL query to aggregate data and response to user. Make sure every table has schema prefix to it in sql query which you can get from 'Get DB Schema and Tables List' tool.
Fetch all data to analyse it for response if needed.

Tools

- Execute SQL query- Executes any sql query generated by AI
- Get DB Schema and Tables List - Lists all the tables in database with its schema name
- Get Table Definition - Gets the table definition from db using table name and schema name"

Execute SQL Query node:

- Tool Description→Set Manually
- Description:
Get all the data from Postgres, make sure you append the tables with correct schema.
Every table is associated with some schema in the database.
- Operation→Execute Query

```
{{ $fromAI("sql_query", "SQL Query") }}
```

Meaning of this expression is as below:

◆ AI Overview हिन्दी 🔊

In n8n, the expression `{{ $fromAI("sql_query", "SQL Query") }}` is a special, dynamic function used within AI-powered workflows to extract a specific piece of information generated by an AI Agent (usually an LLM like GPT-4) and use it within a subsequent node.

Here is a breakdown of what this specifically means:

- `$fromAI(...)`** : This is a method designed for AI agent tools to retrieve data generated by the LLM that is not part of the final text output, but rather a structured output intended for tool execution.
- `"sql_query"`** : This is the **key** or variable name that the AI Agent was instructed to generate. It tells the expression which part of the AI's output to look for.
- `"SQL Query"`** : This is a human-readable description for the variable, often used for debugging or documentation within the node parameters. 

Context of Use: Database Agents

This expression is almost always used in the **Execute SQL Query** node (a PostgreSQL or MySQL tool) within an AI Agent workflow.

- User Input:** The user asks a question in natural language, e.g., *"How many users signed up today?"*.
- AI Agent:** The LLM interprets the question and generates a JSON object containing the SQL query, e.g.,

```
{ "sql_query": "SELECT count(*) FROM users WHERE created_at = '...';" }
```

.
- `$fromAI` Function:** The SQL node uses `{{ $fromAI("sql_query", "SQL Query") }}` to extract just the SQL string (`SELECT ...`) from that JSON object and execute it against the database.

Get DB Schema node

- Tool Description→Set Manually
- Description:
Get list of all tables with their schema in the database.
- Operation→Execute Query

```
SELECT
```

```

        table_schema,
        table_name
FROM information_schema.tables
WHERE table_type = 'BASE TABLE'
      AND table_schema NOT IN ('pg_catalog',
                              'information_schema')
ORDER BY table_schema, table_name;

```

Get Table Definition

- d. Tool Description→Set Manually
- e. Description:
Get table definition to find all columns and types
- f. Operation→Execute Query

```

select
    c.column_name,
    c.data_type,
    c.is_nullable,
    c.column_default,
    tc.constraint_type,
    ccu.table_name AS referenced_table,
    ccu.column_name AS referenced_column
from
    information_schema.columns c
LEFT join
    information_schema.key_column_usage kcu
    ON c.table_name = kcu.table_name
    AND c.column_name = kcu.column_name
LEFT join
    information_schema.table_constraints tc
    ON kcu.constraint_name = tc.constraint_name
    AND tc.constraint_type = 'FOREIGN KEY'
LEFT join
    information_schema.constraint_column_usage ccu
    ON tc.constraint_name = ccu.constraint_name
where
    c.table_name = '{{ $fromAI("table_name") }}'
    AND c.table_schema = '{{ $fromAI("schema_name") }}'
order by
    c.ordinal_position

```

Create tables

(See [this file](#) (at line no 170-211)for all the quick code):

```

CREATE TABLE departments (
    department_id int PRIMARY KEY,
    department_name VARCHAR(100) NOT NULL
);

CREATE TABLE employees (
    employee_id int PRIMARY KEY,
    employee_name VARCHAR(100) NOT NULL,
    age float,
    departmentId INT REFERENCES departments (department_id)
);

```

INSERT Data:

```
INSERT INTO departments (department_id,department_name) values
(1, 'Human Resources'),
(10,'Finance'),
(20,'IT'),
(30, 'Operations') ;
```

```
INSERT INTO employees (employee_id,employee_name, age, departmentId) values
(1,'Alice', 30.4, 10),
(2,'Bob',50.8, 10),
(3,'sudha', 39.0, 20),
(4,'Charlie',34.0, 30);
```

- G. Data transformation
- H. Sub-workflows in n8n
- I. Using webhook in n8n
- J. Slack

How to get slack api or token from slack. Here are the steps:

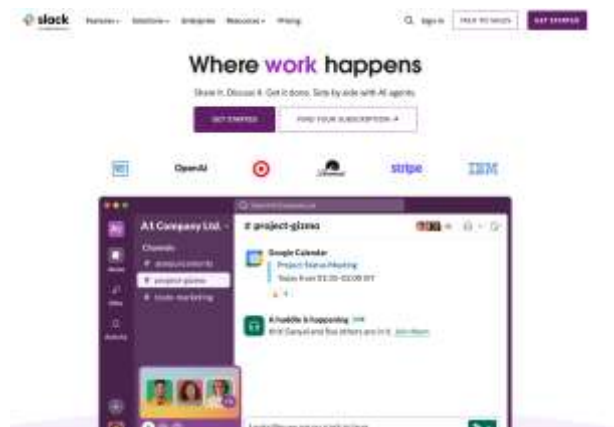


Figure 49: Sign in using your email.



Figure 50: Create a new Workspace of your own where your team will work

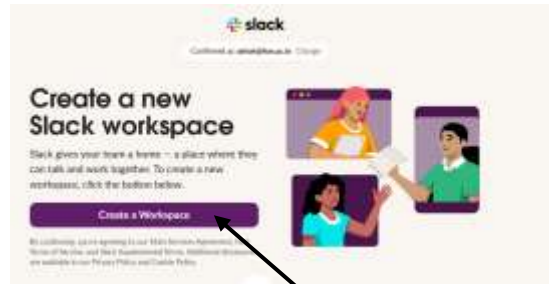


Figure 51: Create a Workspace

The image shows the Slack 'What's the name of your company or team?' screen. At the top, it says 'Step 1 of 4'. The main heading is 'What's the name of your company or team?'. Below this, a paragraph explains that this will be the name of your Slack workspace — choose something that your team will recognize. There is a text input field with 'FORE School' entered. Below the input field, there is a checkbox labeled 'Let anyone with an @forensics.in email join this workspace.' which is checked. At the bottom, there is a purple button labeled 'Next'.

Figure 52: Fill in some details about your workspace. Name your team and click Next

The image shows the Slack 'Who else is on the FORE School team?' screen. At the top, it says 'Step 2 of 4'. The main heading is 'Who else is on the FORE School team?'. Below this, there is a link 'Add members by email' and a link 'Add from Google Contacts'. There is a text input field with 'csm@forensics.in' entered. Below the input field, there is a checkbox labeled 'Keep in mind that invitations expire in 50 days. You can always extend that deadline.' which is checked. At the bottom, there is a purple button labeled 'Next', a link 'Copy Invite Link', and a link 'Skip this step'. An arrow points to the 'csm@forensics.in' text.

Figure 53: Add team members

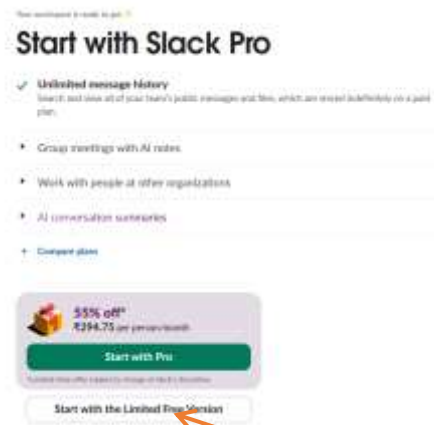


Figure 54: No. Start with a limited free version

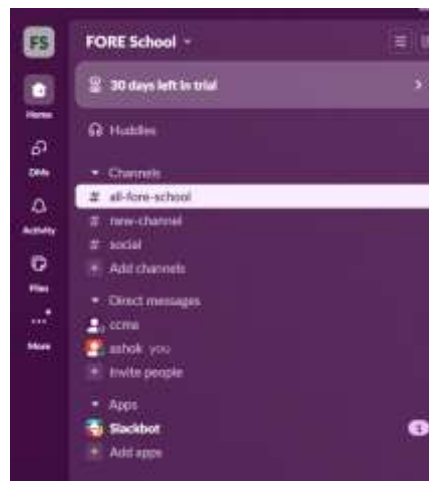


Figure 55: Three channels, one app

What are Slack channels:

A Slack channel is a dedicated space within a Slack workspace where teams can collaborate, communicate, and share information related to a specific topic, project, or team. Channels can be public, allowing anyone in the workspace to join, or private, requiring explicit invitation. Key features of channels are:

Collaboration: Channels facilitate communication, file sharing, and even video/audio calls.

Searchability: Messages and files within a channel are searchable, making it easy to find past conversations and information.

Customization: Channels can be created for various purposes, such as project updates, team discussions, or specific topics.

A company might have a public channel for general announcements (#general), a private channel for sensitive HR matters (#hr-private), and a project-specific channel for a new product launch (#product-launch).

What is an app

Apps connect other software that you use (such as Google Calendar, OneDrive or one of your company's internal tools) to Slack. With all your tools in one place, you can streamline work and help people in your workspace collaborate more effectively.

What you need to know

- There are a few different types of apps that you may see in Slack – built by Slack, third parties or your own team. How an app was built determines how it can be installed and managed in a workspace, as well as where and how you'll be able to interact with it.
- By default, any workspace member can [install apps](#), but owners and admins can choose to [restrict this permission](#). Once an app is installed to a workspace, any member can connect their account to use it.
- Before installing an app from the Slack Marketplace, you can review its privacy policy and security and compliance information (if submitted by the app's developer) from the app page. We recommend only choosing services that you trust when installing apps to Slack.

K. Pinecone vector store:

1. Create a free account (say, using google)
2. Create an API key
3. Create a (vector) index

Understanding Pinecone indexes:

In Pinecone, an index is the primary organizational unit for storing and querying vector data. Think of it as a table in a database, but specifically designed for efficient similarity searches on vectors.

Indexes can accept, store, and serve queries on vectors, as well as perform other vector operations.

Here's a more detailed breakdown:

Storage and Organization:

Indexes hold the vector embeddings of your data, allowing you to store and manage them in a structured way.

Key Parameters dimension etc:

When creating an index, you'll need to define its name, the dimensionality of the vectors it will store, and the similarity metric (e.g., cosine, Euclidean).

Types of Indexes for scaling out:

Pinecone offers [serverless](#) and [pod-based](#) indexes, with different options for scaling and performance.

Namespaces:

Within an index, you can further organize data using namespaces, allowing you to isolate queries to specific subsets of your data.

Similarity Search:

Pinecone indexes are built for efficient similarity searches, meaning you can quickly find vectors that are similar to a given query vector.

Highest Level:

It's the top-level container for your vector data within Pinecone, similar to a table in a relational database.

Creation and Management:

Indexes can be created via the Pinecone UI or programmatically using their API.

Metadata:

Pinecone indexes can store associated metadata with each vector, enabling filtering and more complex search conditions.

Metadata in Pinecone

Every [record](#) in an index must contain an ID and a vector. In addition, you can include metadata key-value pairs to store additional information or context. When you query the index, you can then include a [metadata filter](#) to limit the search to records matching a filter expression. Searches without metadata filters do not consider metadata and search the entire namespace.

See [LlamaIndex example](#) for Chromadb as to how metadata filter works.

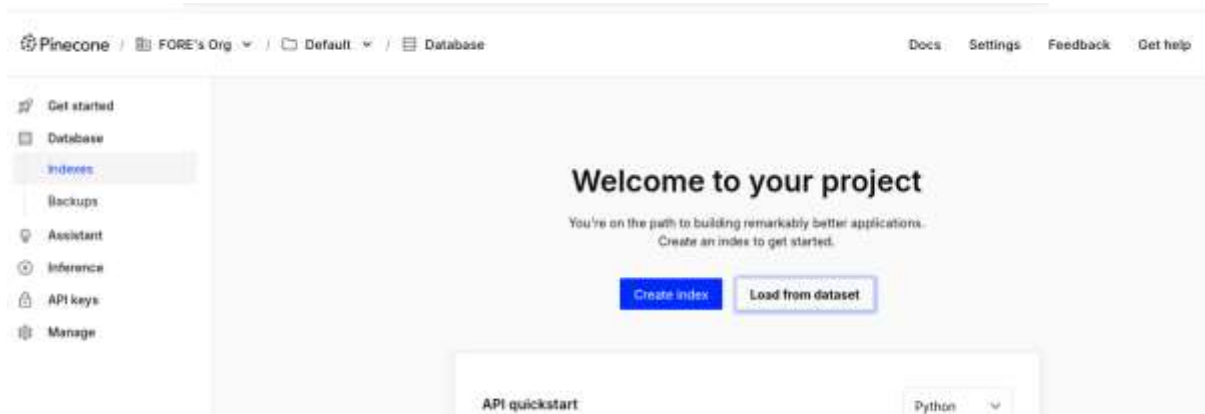


Figure 56: Click Create Index to begin creating an index

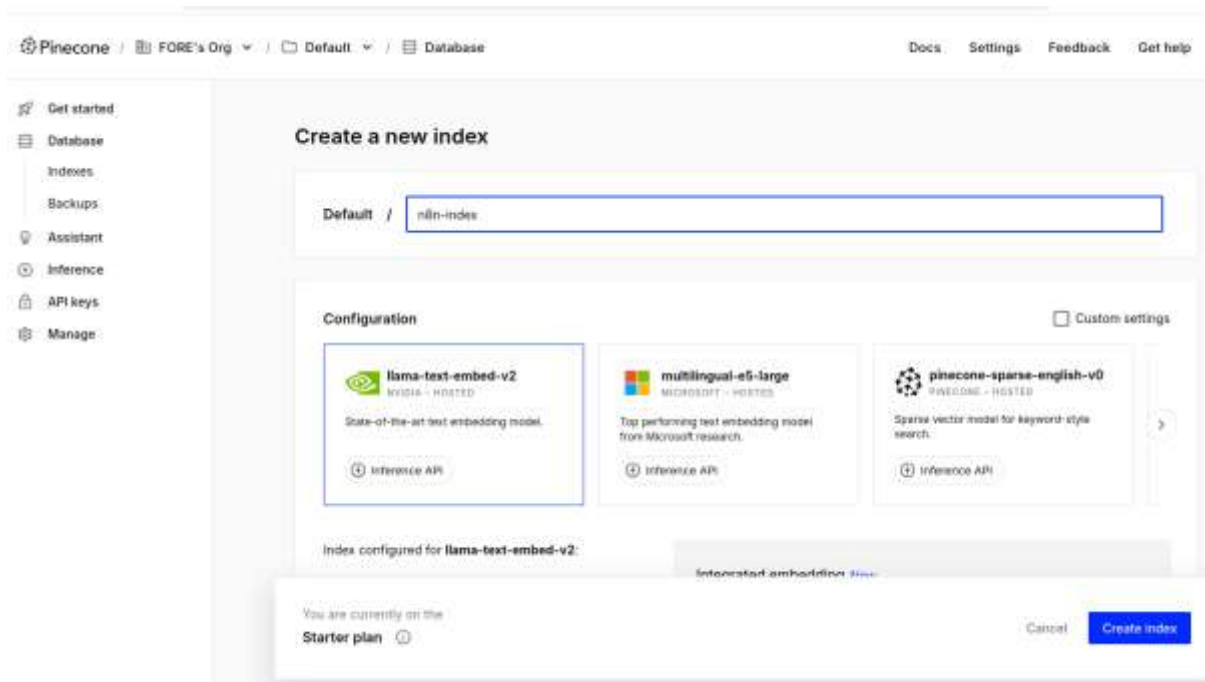


Figure 57: Select an index and select a Configuration for index (such as Vector Dimension: 768, 1024 etc). See figure below

Configuration

**llama-text-embed-v2**
NVIDIA - HOSTED

State-of-the-art text embedding model.

 Inference API

Index configured for **llama-text-embed-v2**:

Modality	Text
Vector type	Dense
Max input	2,048 tokens
Starter limits	5M tokens
Dimension	1024 
Metric	cosine

Figure 58: Set vector Dimension as per your embedder (click Down-arrow). For example, nomic-embed-text has a vector dimension of 768 AND NOT of 1024. We have used 768

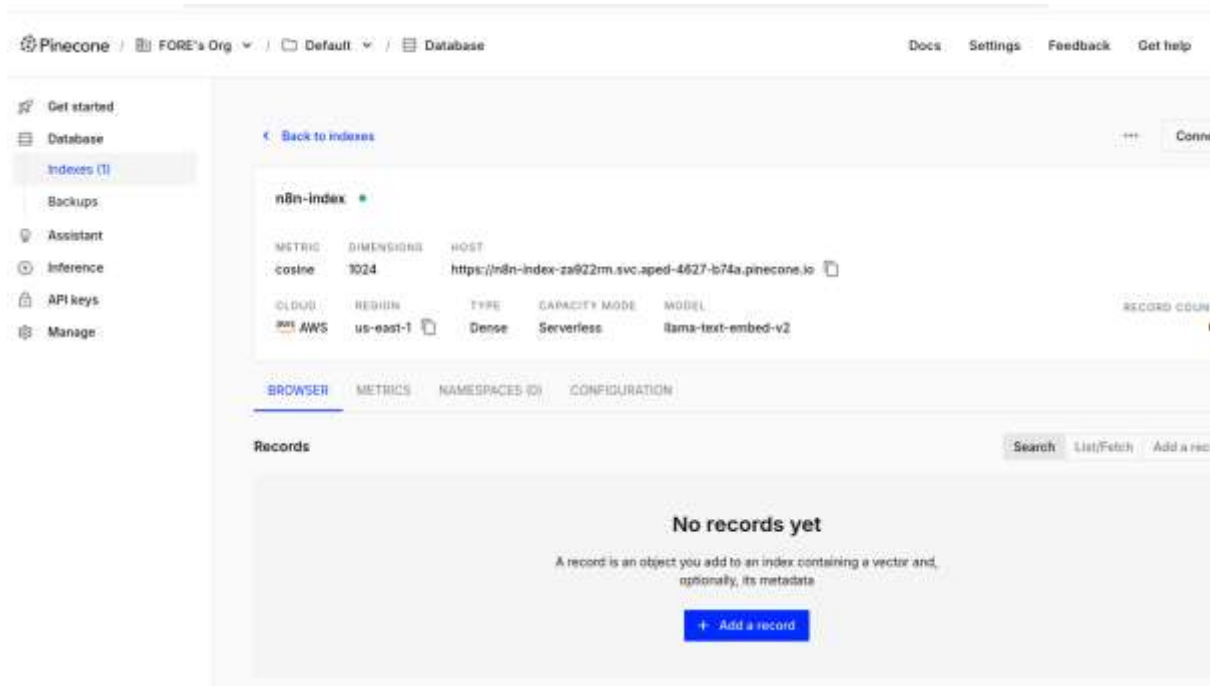


Figure 59: Create index. Records will come from n8n workflow

Note that selected embedder will also decide max input tokens. The character splitter chunk size will affect the amount of **input tokens** you're trying to get. For the embedding model of *mxlbai-embed-large* (vector dimension: 1024), max input token limit is 512.

Our dataset is *students.json* on GitHub, [at site](#)

```
{
  "_id": 0, "name": "aimee Zank", "scores": [
    { "score": 1.463179736705023, "type": "exam" },
    { "score": 11.78273309957772, "type": "quiz" },
    { "score": 35.8740349954354, "type": "homework" }
  ]
}, {
  "_id": 1, "name": "Aurelia Menendez", "scores": [
    { "score": 60.06045071030959, "type": "exam" },
    { "score": 52.79790691903873, "type": "quiz" },
    { "score": 71.76133439165544, "type": "homework" }
  ]
}, {
  "_id": 2, "name": "Corliss Zuk", "scores": [
    { "score": 67.03077095065002, "type": "exam" },
    { "score": 6.301851677035235, "type": "quiz" },
    { "score": 66.28344683278382, "type": "homework" }
  ]
}, {
  "_id": 3, "name": "Bao Ziglar", "scores": [
    { "score": 71.64343899778332, "type": "exam" },
    { "score": 24.80221293650313, "type": "quiz" },
    { "score": 42.26347058804812, "type": "homework" }
  ]
}, {
  "_id": 4, "name": "Zachary Langlais", "scores": [
    { "score": 78.68385091384332, "type": "exam" },
    { "score": 90.2963101368042, "type": "quiz" },
    { "score": 34.41620148842529, "type": "homework" }
  ]
}, {
  "_id": 5, "name": "Wilburn Spiess", "scores": [
    { "score": 44.87186336181261, "type": "exam" },
    { "score": 25.72395114668016, "type": "quiz" },
    { "score": 63.42288310628662, "type": "homework" }
  ]
}, {
  "_id": 6, "name": "Janette Flanders", "scores": [
    { "score": 37.32285459166097, "type": "exam" },
    { "score": 28.32634976913737, "type": "quiz" },
    { "score": 81.57115318686338, "type": "homework" }
  ]
}, {
  "_id": 7, "name": "Salma Olmos", "scores": [
    { "score": 90.37826509157176, "type": "exam" },
    { "score": 42.48780666956811, "type": "quiz" },
    { "score": 96.5298617163331, "type": "homework" }
  ]
}, {
  "_id": 8, "name": "Daphne Zhong", "scores": [
    { "score": 22.13583712862635, "type": "exam" },
    { "score": 14.63869941335869, "type": "quiz" },
    { "score": 75.94123677556644, "type": "homework" }
  ]
}
```

Figure 60: Extract from file *students.json*

Here are two rows from the data:

```
{ "_id": 0, "name": "aimee Zank", "scores": [
  { "score": 1.463179736705023, "type": "exam" },
  { "score": 11.78273309957772, "type": "quiz" },
  { "score": 35.8740349954354, "type": "homework" }
] }

{ "_id": 1, "name": "Aurelia Menendez", "scores": [
  { "score": 60.06045071030959, "type": "exam" },
  { "score": 52.79790691903873, "type": "quiz" },
  { "score": 71.76133439165544, "type": "homework" }
] }
```

L. Simple RAG with n8n

File: 'Simple RAG flow.json'

We use '*n8n Form trigger*' node as file uploader. The vector store is in-memory vector store: Simple Vector Store. File used is: *bertrandRusselEssays.txt*

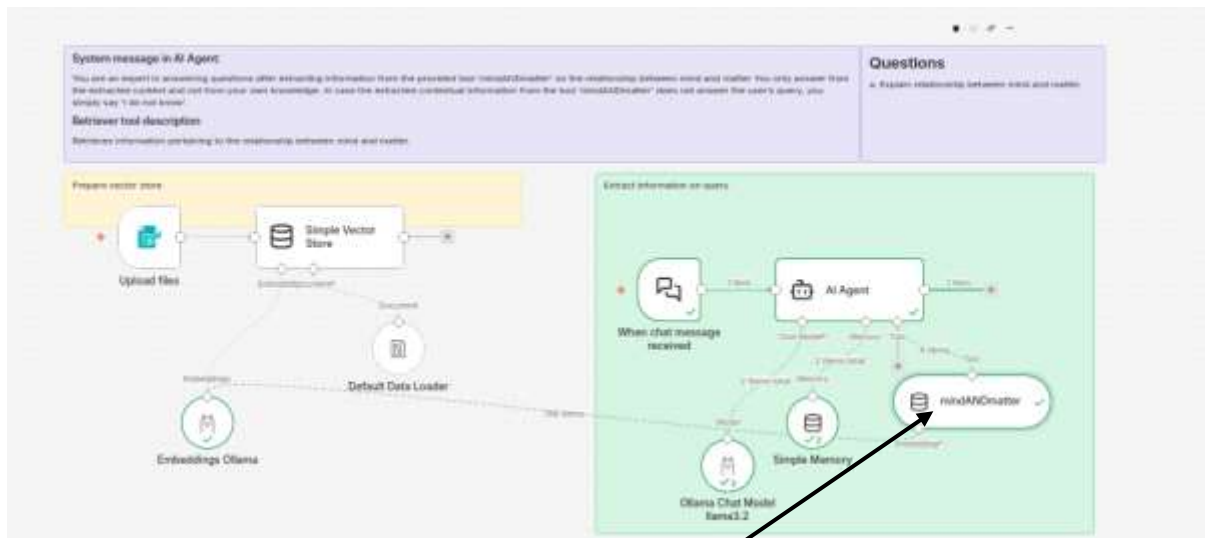


Figure 61: Full workflow. NOTE that even though this is also Simple Vector Store but has been renamed as per the subject.

Messages are as follows:

System Message:

You are an expert in answering questions after extracting information from the provided tool 'mindANDmatter' on the relationship between mind and matter. You only answer from the extracted context and not from your own knowledge. In case the extracted contextual information from the tool 'mindANDmatter' does not answer the user's query, you simply say 'I do not know'.

Tool Des:

Retrieves information pertaining to the relationship between mind and matter.

Its two sub-workflows are as follows:

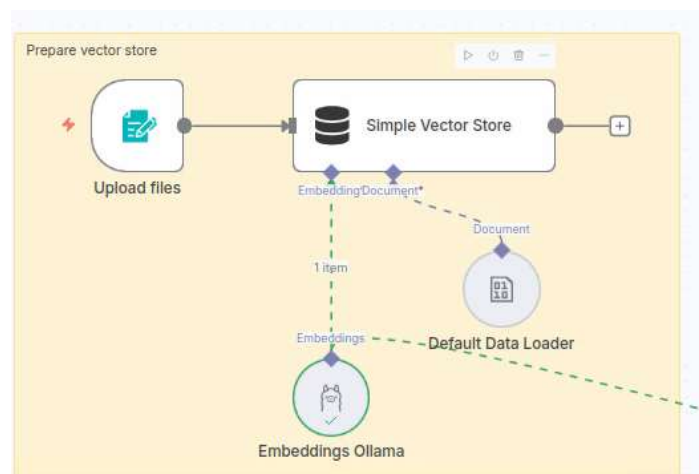


Figure 62: Creating a vector store

Upload files node (n8n form):

The screenshot shows the configuration interface for the 'Upload files' node in n8n. The interface is dark-themed and includes a top bar with 'Parameters' and 'Settings' tabs, and an 'Execute step' button. The main area is titled 'Form Elements' and contains a list of form elements. The first element is a 'File' type, which is expanded to show its configuration. The configuration includes a 'Label' field with the text 'Select your file', an 'Element Type' dropdown set to 'File', a 'Custom Field Name' field with the text 'uploadedFile', a 'Multiple Files' toggle switch that is turned on, an 'Accepted File Types' field with the text '.pdf,.txt', and a 'Required Field' toggle switch that is turned on. There are also buttons for '+ Add Attributes' and '+ Add Form Element'.

Parameters Settings Execute step

Form Elements +

File

Label ⓘ

Select your file

Element Type

File

Custom Field Name

uploadedFile

Multiple Files

Accepted File Types

.pdf,.txt

Leave empty to allow all file types

Required Field

+ Add Attributes

+ Add Form Element

Figure 63: File upload with n8n form. Allow both pdf and txt files upload. Easy to upload Paul Graham essay on Good Writing.

Simple vector store on the left side of workflow

The screenshot shows the configuration interface for a Simple Vector store. It has two tabs: 'Parameters' (selected) and 'Settings'. A red 'Execute step' button is in the top right. A tip box at the top says: 'Tip: Get a feel for vector stores in n8n with our [RAG starter template](#)'. Below this, the 'Operation Mode' dropdown is set to 'Insert Documents'. The 'Memory Key' section has a 'From list' dropdown and a text input field containing 'vector_store_key'. The 'Embedding Batch Size' is a text input field with the value '200'. At the bottom, there is a 'Clear Store' toggle switch which is currently turned off.

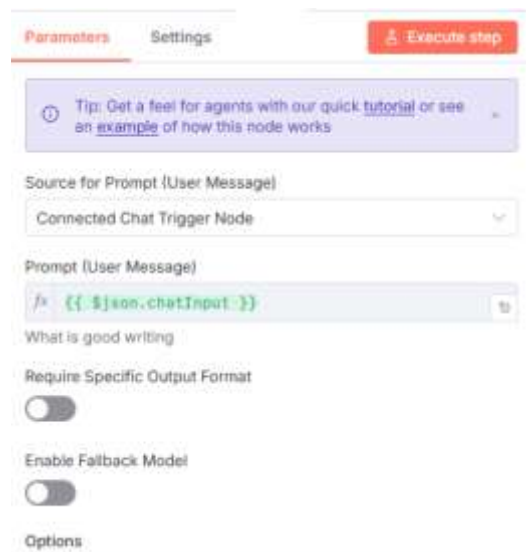
Figure 64: Simple Vector store configuration

Default Data loader

The screenshot shows the configuration interface for the Default Data loader. It has two tabs: 'Parameters' (selected) and 'Settings'. An orange tip box at the top says: 'This will load data from a previous step in the workflow. Example'. Below this, the 'Type of Data' dropdown is set to 'Binary'. The 'Mode' dropdown is set to 'Load All Input Data'. The 'Data Format' dropdown is set to 'Automatically Detect by Mime Type'. The 'Text Splitting' dropdown is set to 'Simple'. At the bottom, there is an 'Options' section which currently displays 'No properties'.

Figure 65: Default data loader. Type of data is Binary (the other choice is JSON).. Text splitting is Simple

AI Agent has no special system prompt. Default is OK. But you should change. See before the suggested system message.



Parameters Settings Execute step

Tip: Get a feel for agents with our quick [tutorial](#) or see an [example](#) of how this node works

Source for Prompt (User Message)
Connected Chat Trigger Node

Prompt (User Message)
/{{ \$json.chatInput }}

What is good writing

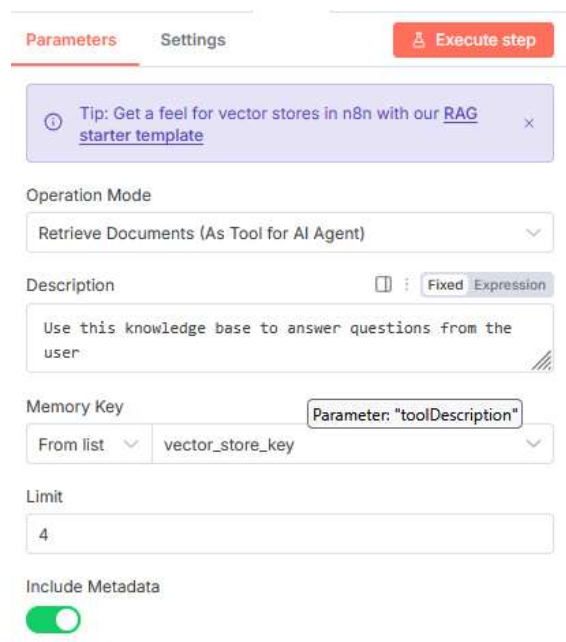
Require Specific Output Format
☐

Enable Fallback Model
☐

Options

Figure 66: No special System prompt in AI Agent

Simple vector store below AI Agent. Should be named properly.



Parameters Settings Execute step

Tip: Get a feel for vector stores in n8n with our [RAG starter template](#)

Operation Mode
Retrieve Documents (As Tool for AI Agent)

Description
Fixed Expression
Use this knowledge base to answer questions from the user

Memory Key
From list vector_store_key Parameter: "toolDescription"

Limit
4

Include Metadata
☒

Figure 67: Simple vector store on AI Agent side

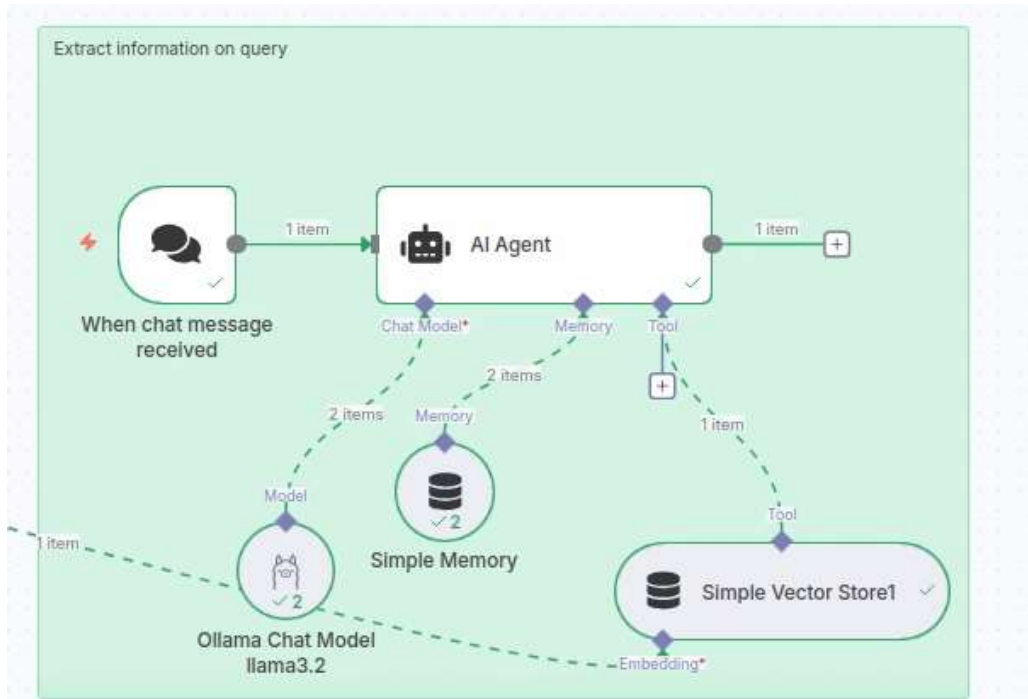


Figure 68: Querying a vector store

System message of AI Agent and vector tool description are as follows:

System message in AI Agent:
 You are an expert in answering questions after extracting information from the provided 'Simple Vector Store1' on the game of football. You only answer from the extracted context and not from your own knowledge. In case the extracted contextual information from 'Simple Vector Store1' does not answer the user's query, you simply say 'I do not know'.

Retriever tool description:
 Retrieves information pertaining to the game of football

Figure 69: System message for AI Agent

M. RAG with Qdrant

This works better than the above.. See [this link](#). Quick two lines code to install qdrant(copy and paste):

```
# 1
mkdir -p ~/qdrant_storage

# 2.
docker run -d -p 6333:6333 -p 6334:6334 -v ~/qdrant_storage:/qdrant/storage --name
my_qdrant qdrant/qdrant

# 3
Next time start as:
docker start my_qdrant
```

Then create the following two workflows.

Hints:

- 'Default Data Loader' must have 'Binary' set rather than the default 'Json'.
- Give a proper System Message in AI Agent. System message mentions the name of tool to be used ('retriever', for example) and when to be used. (You are a helpful assistant. When any question is asked get its answer from 'retriever' tool. Under no circumstances answer a question or add to it from your own knowledgebase--only use the 'retriever' tool to answer it.)
- Name the Qdrant vector tool also as: 'retriever'.
- Use `mxbai-embed-large` OR `bge-m3` (vector dimension: 1024) as embedding model and chunk size as: 200/20
- Use Ollama cloud models. To use ollama cloud models, first execute 'ollama signin'; then go to the web-page and sign into it and Connect your device. Next, in the console execute 'ollama run minimax-m3:cloud'. Then say, /bye. From now on, this model will appear in 'ollama list'.

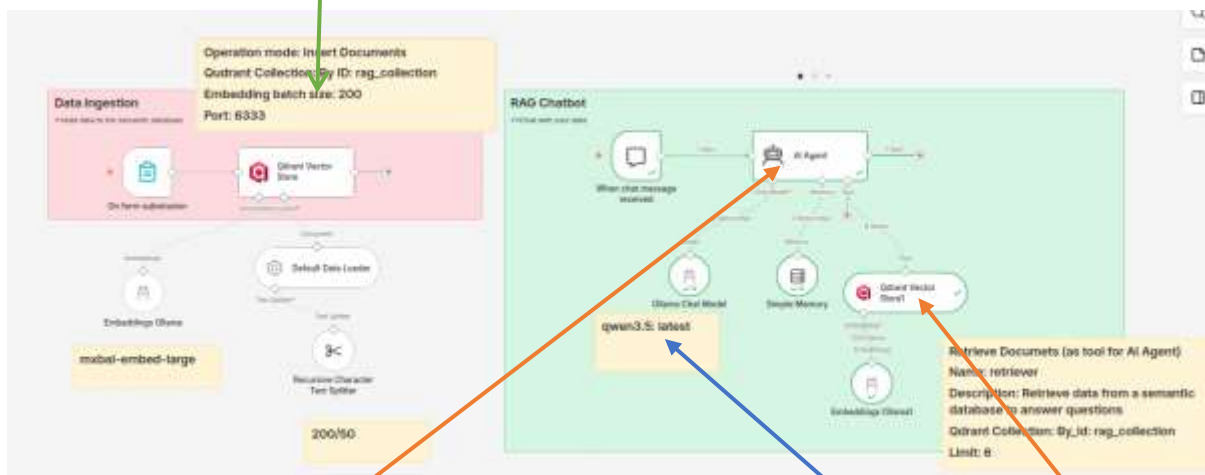


Figure 70: Write a proper System Message. The Collection ID name should be the same in both the branches. Also give a proper name to Retriever tool (such as: 'retriever'). Or, use a small ollama cloud model: `minimax-m3:cloud`; `llama3.2` also works.

The screenshot shows the 'Parameters' tab of a configuration interface. The 'Operation Mode' is set to 'Retrieve Documents (As Tool for AI Agent)'. The 'Description' is 'Answer questions related to BrowserStack'. The 'Qdrant Collection' is set to 'By ID' with the value 'abcd'. The 'Limit' is set to 4. The 'Include Metadata' checkbox is checked. The 'Rerank Results' checkbox is unchecked. The 'Options' section is empty, with a '+ Add Option' button at the bottom.

Figure 71: Retriever is, in fact, Qdrant vector store but with option selected as above.

Here use mxbai-embed-large embedding model

Key Differences Comparison		
Feature	mxbai-embed-large	homic-embed-text (v2)
Dimensions	1,024	768
Context Window	512 tokens	8,192 tokens
Parameters	~334 Million	~137 Million
Model Size	~670 MB	~274 MB
Best For	High accuracy, RAG with smaller chunks	Fast, local, long document indexing
MTEB (Overall)	Higher (~64)	High (~62)

Figure 72:Use mxbai-embed-large embedding model in the above experiment. Keep chunk size low (200/50).

N. RAG with Pinecone

N8n model files: n8n/2025/Rag Flow-I.json and Revised rag flow-II.json
Dataset file: *students.json* OR see [Try this](#). For json data, different chunking strategies can be used. See google.

Using Pinecone
Log-into it with google and get API key. Then proceed to create index as follows:

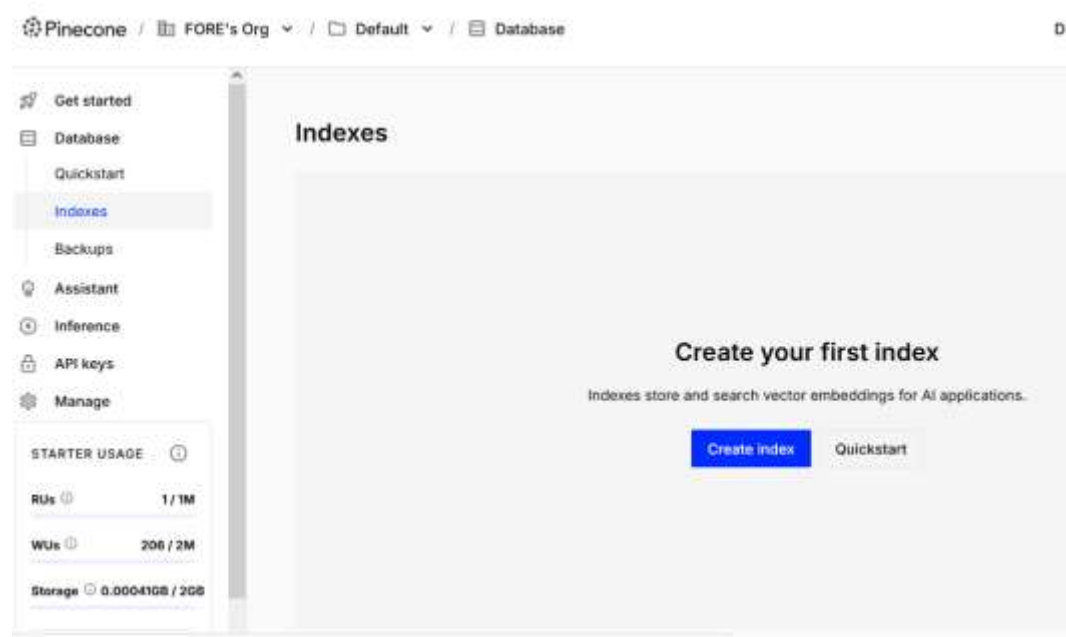


Figure 73: Click Create Index

Figure 74: Create Index with custom settings. As we are using nomic-embed-text, use a dimension of 768

See [this link](#) for the detailed blog. Vectorization may take a very long time depending upon the input size. As vectorization of files may take a long time, a smaller json file is on GitHub [at this link](#). This file has 35 json records.

Experiment in the class with different students using varying chunk size and maybe different embedder also. Check which embedders are good for json files.

bge-m3: 1024 → Gives Nan error

Nomic-embed-text: 768 → Not that good results

mxbai-embed-large: 1024 → Gives good results (Search for 'Bao Ziglar' scores)

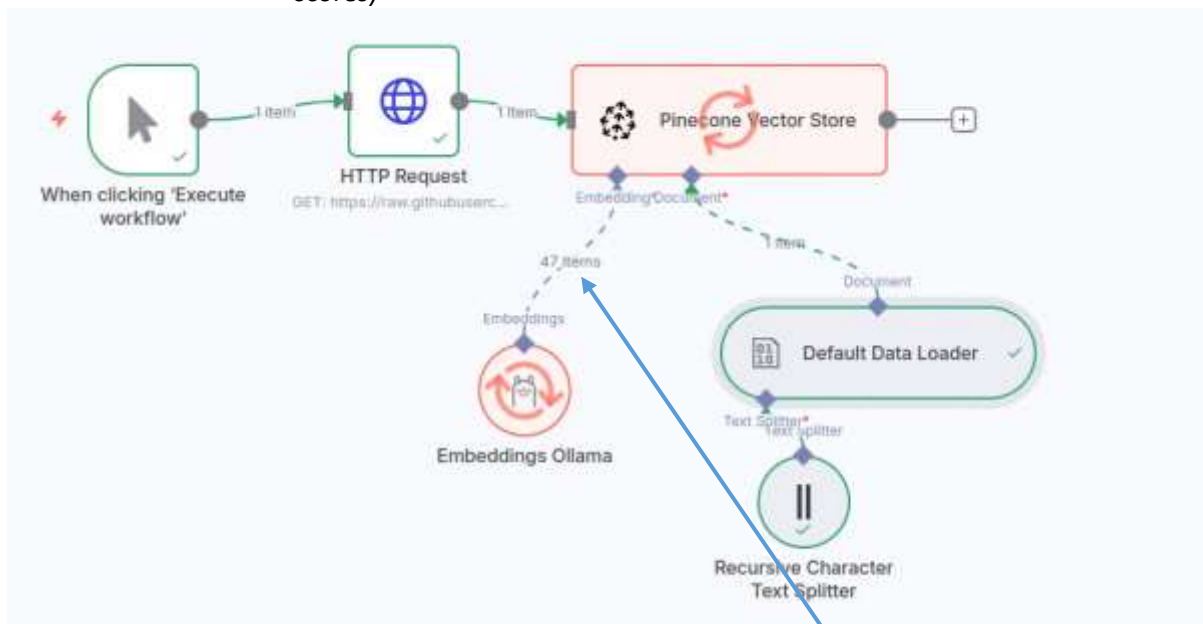


Figure 75: Our workflow to save data to vector store. 47 items possibly mean that 47 chunks were processed till now. Chunk size for Recursive text splitter is 512 as selected embedder is:

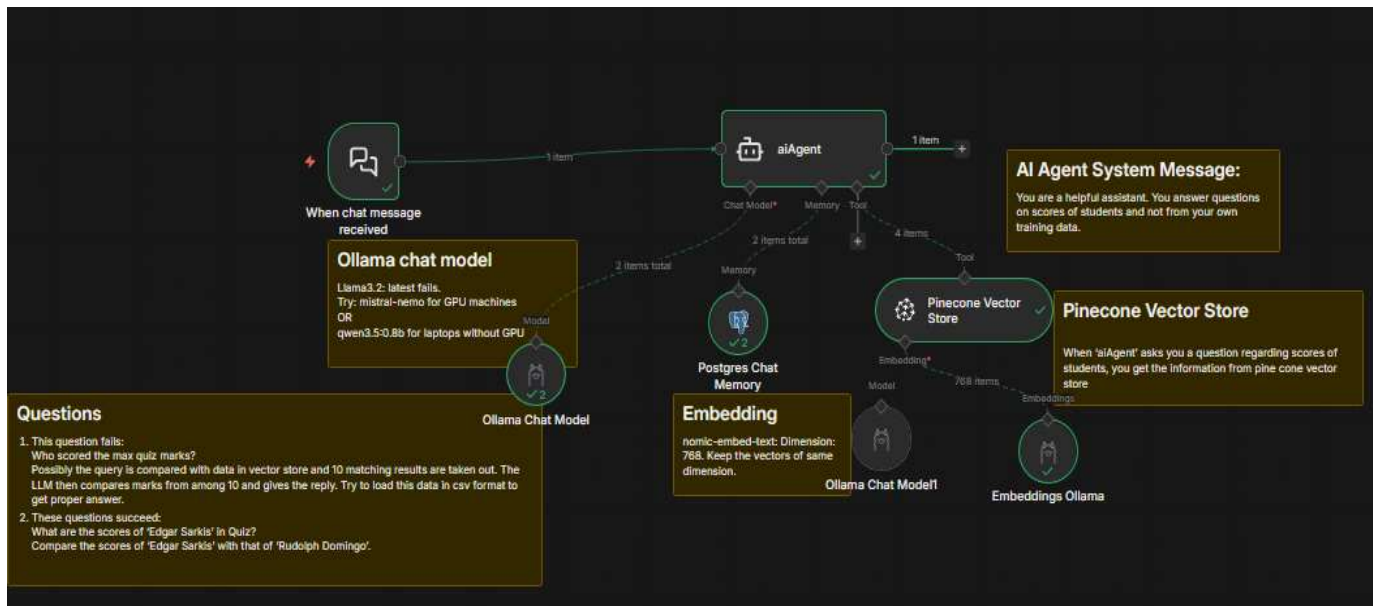
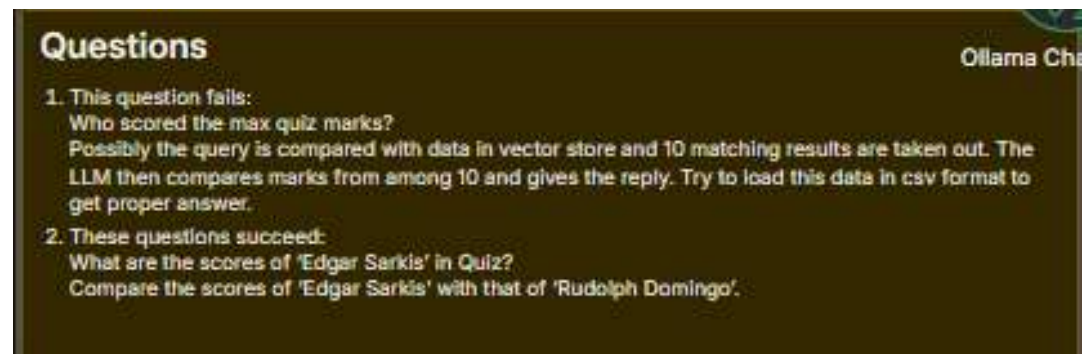


Figure 76: Reading vector data from Pinecone vector store

f. Here are the models used:

Embedding: *nomic-embed-text* (ollama). Dimensions: 768
Chat ollama: *mistral-nemo* (ollama)
qwen3.5:0.8b (for laptops with CPU)

g. Questions asked:



Questions:

- What are the scores of 'Edgar Sarkis' in Quiz?
- Compare the scores of 'Edgar Sarkis' with that of 'Rudolph Domingo'

h. AI Agent System message. NOTE NO TOOL NAME MENTIONED HERE??

AI Agent System Message:

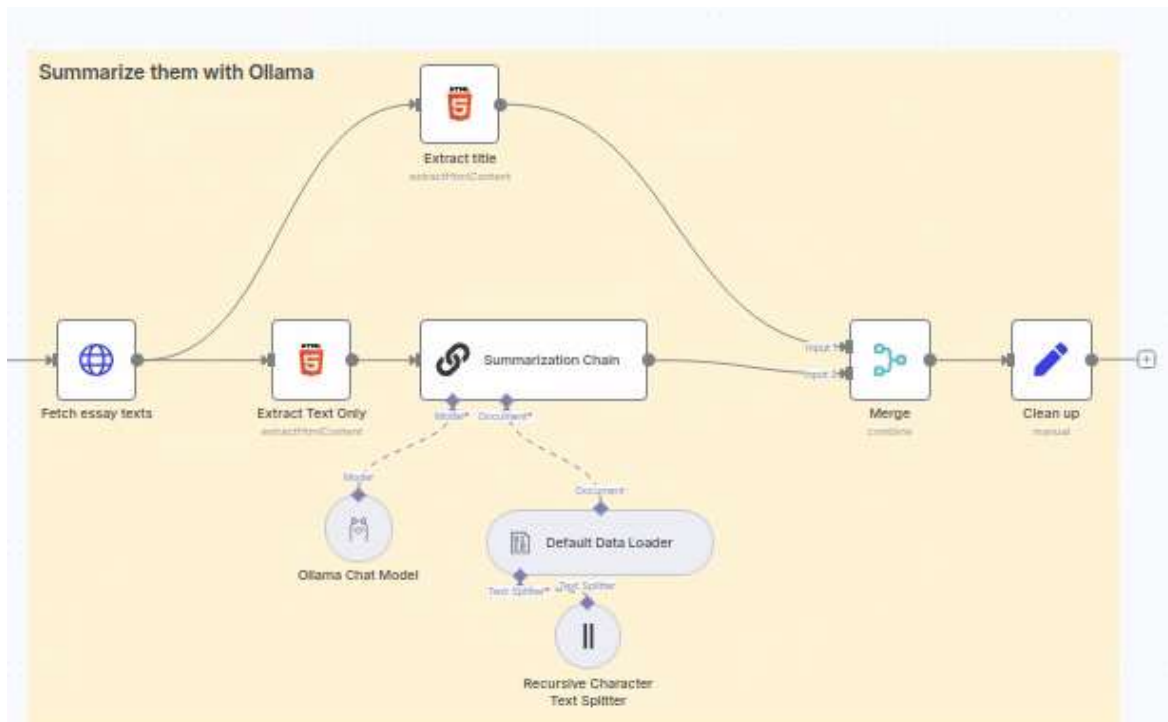
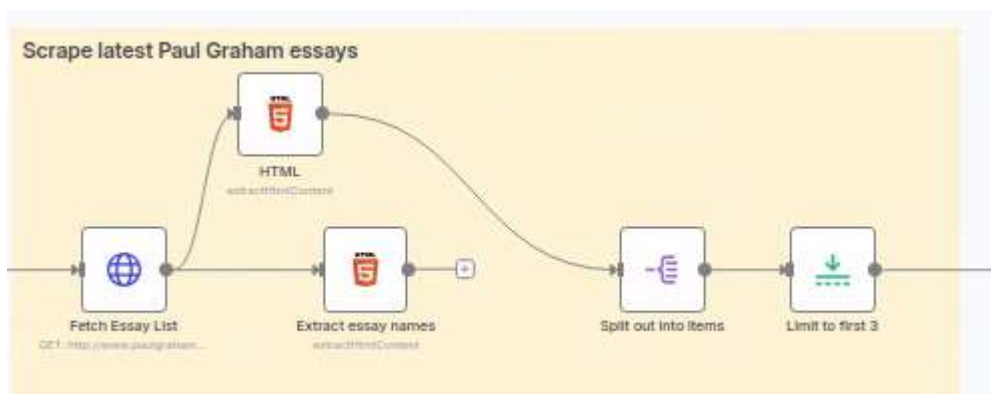
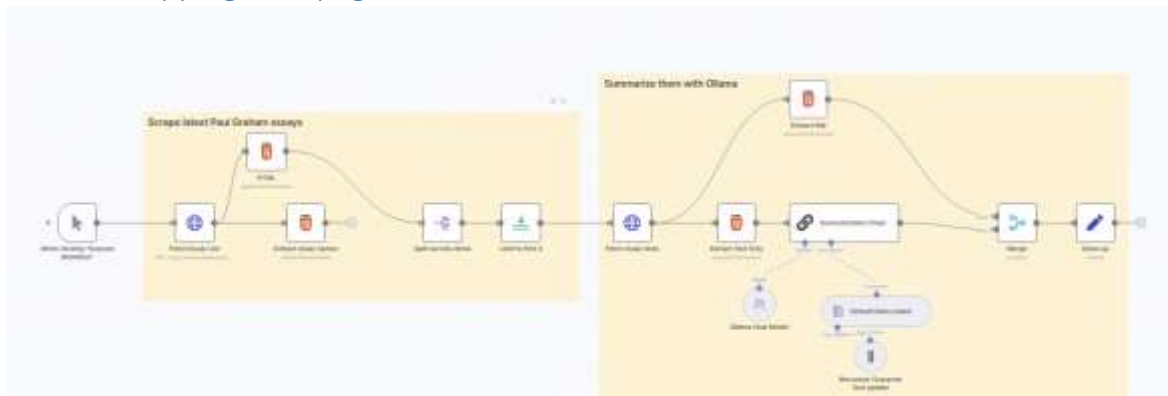
You are a helpful assistant. You answer questions on scores of students and not from your own training data.

i. Pinecone Vector Store Description. TOOL NAME?

Pinecone Vector Store

When 'aiAgent' asks you a question regarding scores of students, you get the information from pine cone vector store

O. Scrapping web-pages and summarization



Fetch Essay List

Execute step

Parameters Settings Docs

Import cURL

Method

GET

URL

http://www.paulgraham.com/articles.html

Authentication

None

Send Query Parameters

Send Headers

Send Body

Options

No properties

Add option

You can view the raw requests this node makes in your browser's developer console

HTML

Parameters Settings

Operation

Extract HTML Content

Source Data

JSON

JSON Property

data

Extraction Values

Key

essay

CSS Selector

table tbody tr td table tbody tr td font a

Return Value

Parameter: "extractionValues.values[0].cssSelector"

Attribute

href

Return Array

Add Value

Split out into items

Execute step

Parameters Settings Docs

Fields To Split Out

essay

Include

No Other Fields

Options

No properties

Add Field

Limit to first 3

Execute step

Parameters Settings Docs

Max Items

3

Keep

First Items

n8n how to decide css selector in html node

Thinking

Kicking off 7 searches

Looking at 105 sites

Putting it all together

Deciding on the correct CSS selector in n8n's HTML node (formerly HTML Extract node) is crucial for accurately extracting the data you need from a web page. Here's a breakdown of how to approach this, including recommended practices and tools:

1. Using Browser Developer Tools (Inspect Element)

The easiest and most common way to identify the correct CSS selector is by using your web browser's developer tools.

- **Navigate to the web page:** Open the web page containing the data you want to scrape in your browser.
- **Inspect the element:** Right-click on the specific element (text, image, link, etc.) you want to extract and select "Inspect" or "Inspect Element". This will open the browser's developer tools and highlight the element's HTML code.
- **Copy the CSS Selector/Path:** In the developer tools (usually under the "Elements" tab), right-click on the highlighted HTML element and choose "Copy" > "Selector". This will copy the CSS selector to your clipboard

Using the CSS selector in the n8n HTML node

- **Add an HTML node to your workflow:** According to n8n Docs and, the node is called "HTML" (it replaced the older "HTML Extract" node).
- **Set Source Data:** Choose "JSON" if the HTML content is in the JSON output of a previous node (like an HTTP Request node), or "Binary" if the HTML is in a binary file.
- **Specify JSON Property (if applicable):** Enter the name of the JSON property containing the HTML.
- **Add Extraction Value:** Click "Add Value" to define what you want to extract.
- **Key:** Enter a name for the extracted value (e.g., "title", "price").
- **CSS Selector:** Paste the CSS selector you identified in the previous steps.
- **Return Value:** Choose the type of data to return: "Attribute" (e.g., "href" for links), "HTML", "Text", or "Value" (for input elements).
- **Return Array (if applicable):** If you expect multiple elements to match the selector, enable "Return Array" to get an array of values.

Example:

If you want to extract the text "Medium" from a `<span>` element within a table, the copied CSS selector might look like:

```
body > table:nth-child(2) > tbody:nth-child(1) > tr:nth-child(1) > td:nth-child(2) > span:nth-child(1)
```

Understanding CSS Selector Syntax:

Familiarize yourself with basic CSS selector syntax (e.g., `.` for classes, `#` for IDs, `>` for direct children, `nth-child()` for selecting elements based on their position) to create or modify selectors manually if needed.

P. Send SMS using Twilio

1. Open [Twilio home](#) page:
2. Click on 'Start for free' button at the top-right
3. At the bottom of this page, click on Sign up with Google

4. After signing in, you have to enter your mobile number so as to verify the destination SMS address. Click the button *Send code via SMS*
5. You will receive a verification code on your mobile by SMS. Enter the code and click Verify button.
6. You will get a recovery code. Copy and Save it somewhere (maybe your email):
XYTMNTY5PGJLS7FA52S5MY3YHP
7. Click *Continue* button
8. Fill up some survey information.
Which channel are you interested in? ==> **Select SMS**
9. You are taken to following screen. This screen has two parts: Top and Bottom.

In the top part, you are invited to get a free telephone number from where you will get an SMS. Note down this number.

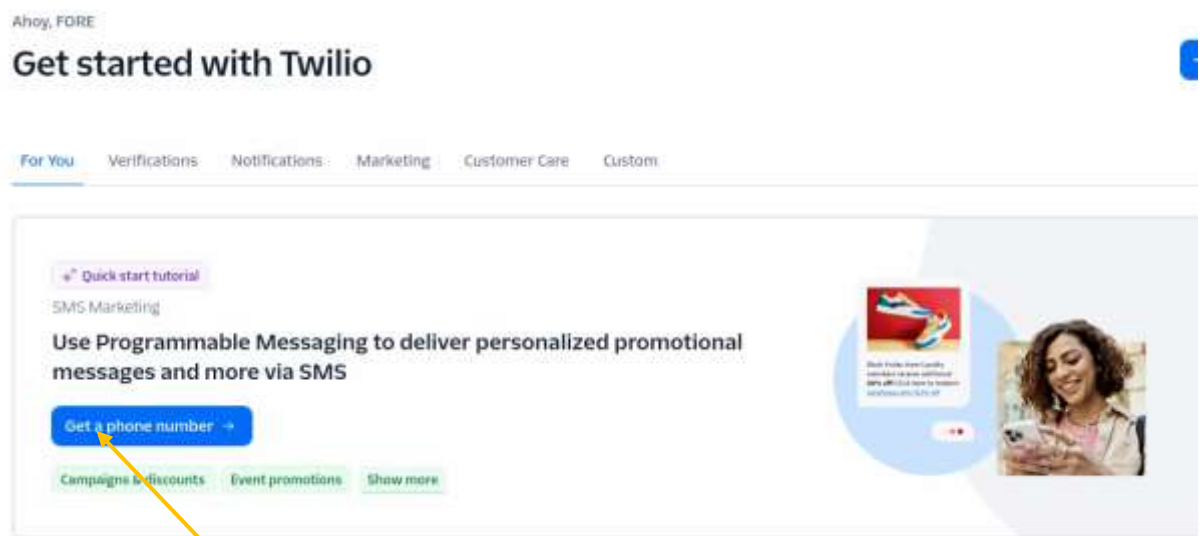


Figure 77: Click the button 'Get Phone Number'

In the bottom part of the page, you get *Account information* and *Auth token*. Save this information also.

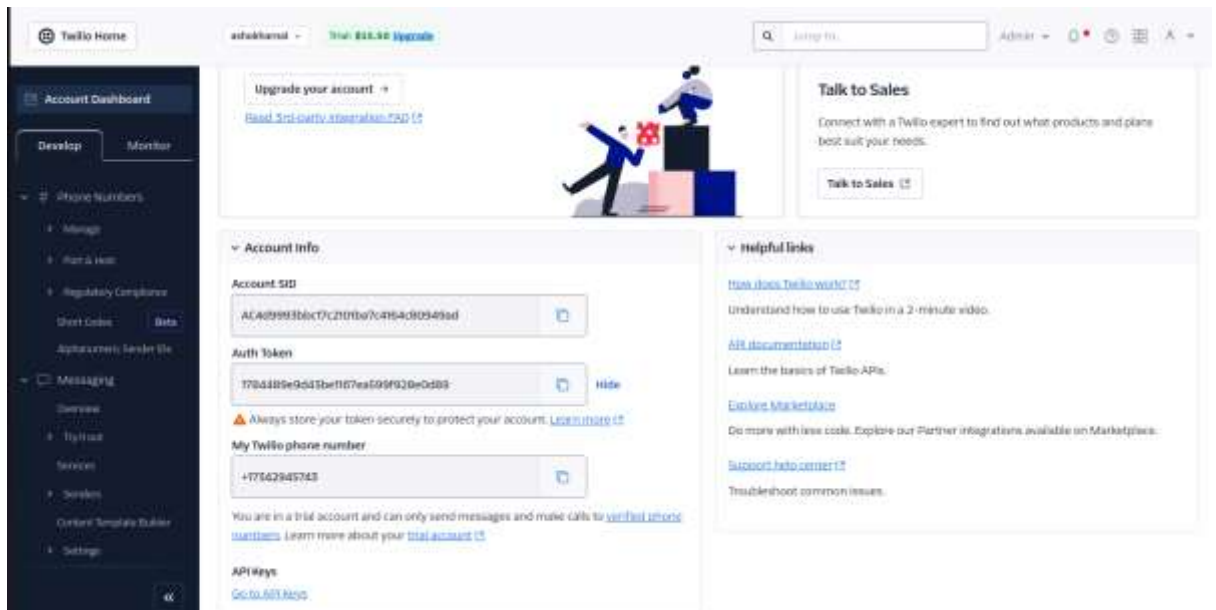


Figure 78: Copy each and every information from this page into your email

Q. Slack API

To work with Slack API, click [on this link](#) or in Google, search for Slack API. [Click here](#) OR [here](#).

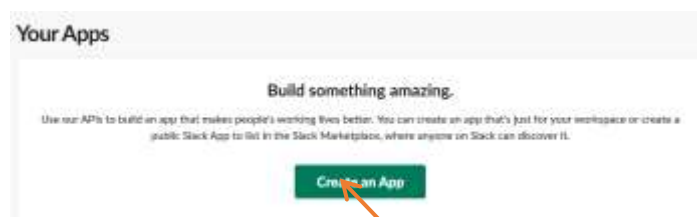


Figure 79: Click on Create an App

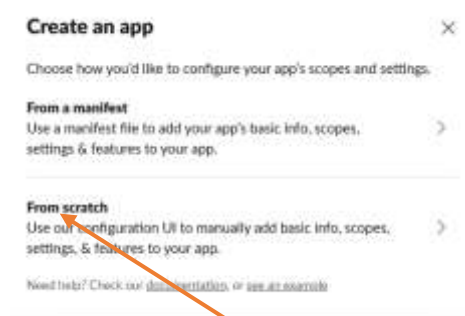


Figure 80: Click From Scratch

Name app & choose workspace

App Name
FORE Service

Don't worry - you'll be able to change this later.

Pick a workspace to develop your app in:
FORE School

Keep in mind that you can't change this app's workspace later. If you leave the workspace, you won't be able to manage any apps you've built for it. The workspace will control the app even if you leave the workspace.

[Switch to a different workspace](#)

By creating a Web API Application, you agree to the [Slack API Terms of Service](#).

Cancel **Create App**

Figure 81: Fill it up and click Create App

Basic Information

App Credentials

These credentials allow your app to access the Slack API. They are secret. Please don't share your app credentials with anyone, include them in public code repositories, or store them in insecure ways.

App ID: A295AG4M2UV Date of App Creation: July 9, 2025

Client ID: 91924404109121912559747984

Client Secret: [REDACTED] [Show](#) [Regenerate](#)

You'll need to send this secret along with your Client ID when making your OAuth 2.0 token request.

Signing Secret: [REDACTED] [Show](#) [Regenerate](#)

Watch sign: The secret is used when you use this secret. Clients that sign requests must use this secret to verify the request's signature.

Verification Token: X0x0K2z0g0y4T7K29Pp04 [Regenerate](#)

This deprecated Verification Token can still be used to verify that requests come from Slack, but we strongly recommend using the above, more secure, signing secret instead.

Figure 82: Not sure if to note down these or not.

FORE Service

Settings

- Basic Information
- Collaborators
- Socket Mode
- Install Apps**
- Manage Distribution

Features

- App Home
- Agents & AI Apps NEW
- Workflow Steps NEW
- Org Level Apps
- Incoming Webhooks
- Interactivity & Shortcuts
- Slash Commands
- Steps from Apps UPDATES
- OAuth & Permissions
- Event Subscriptions
- User ID Transition
- App Manifest NEW
- Beta Features

Submit to Slack Marketplace

- Review & Submit

Figure 83: Click Install Apps

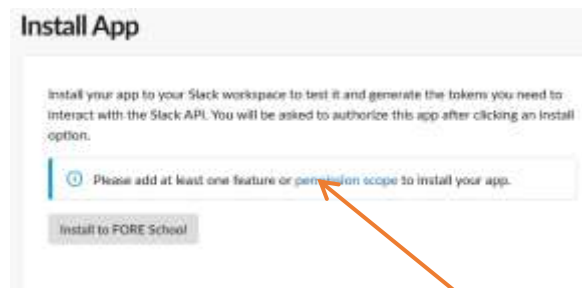


Figure 84: Click Install to FORE School. But first Give permissions



Figure 85: Allow your App permissions on your Workspace

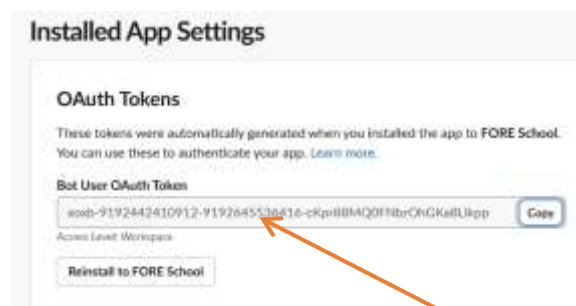


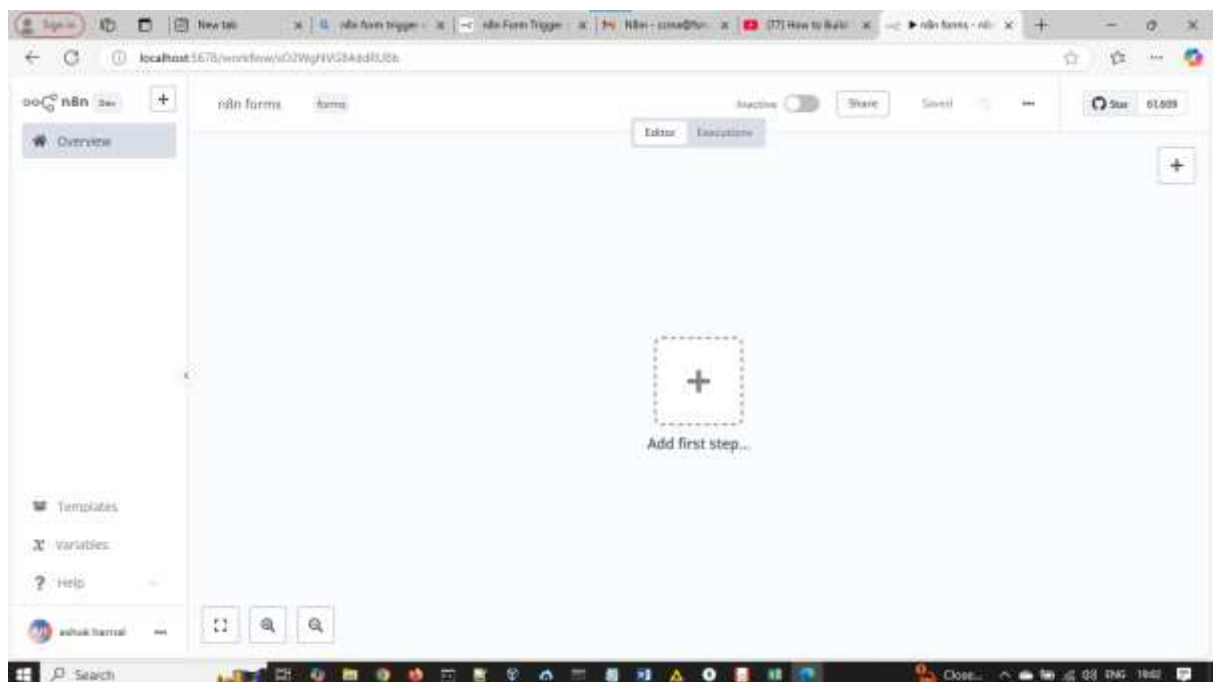
Figure 86: A token will be generated. Note this down. This token is important.

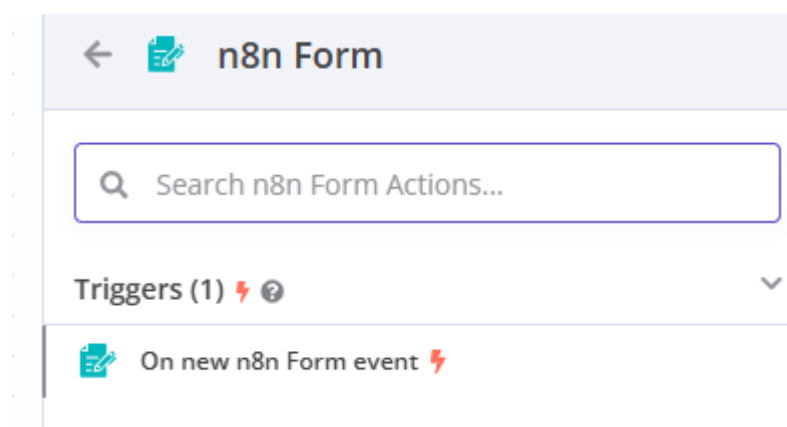
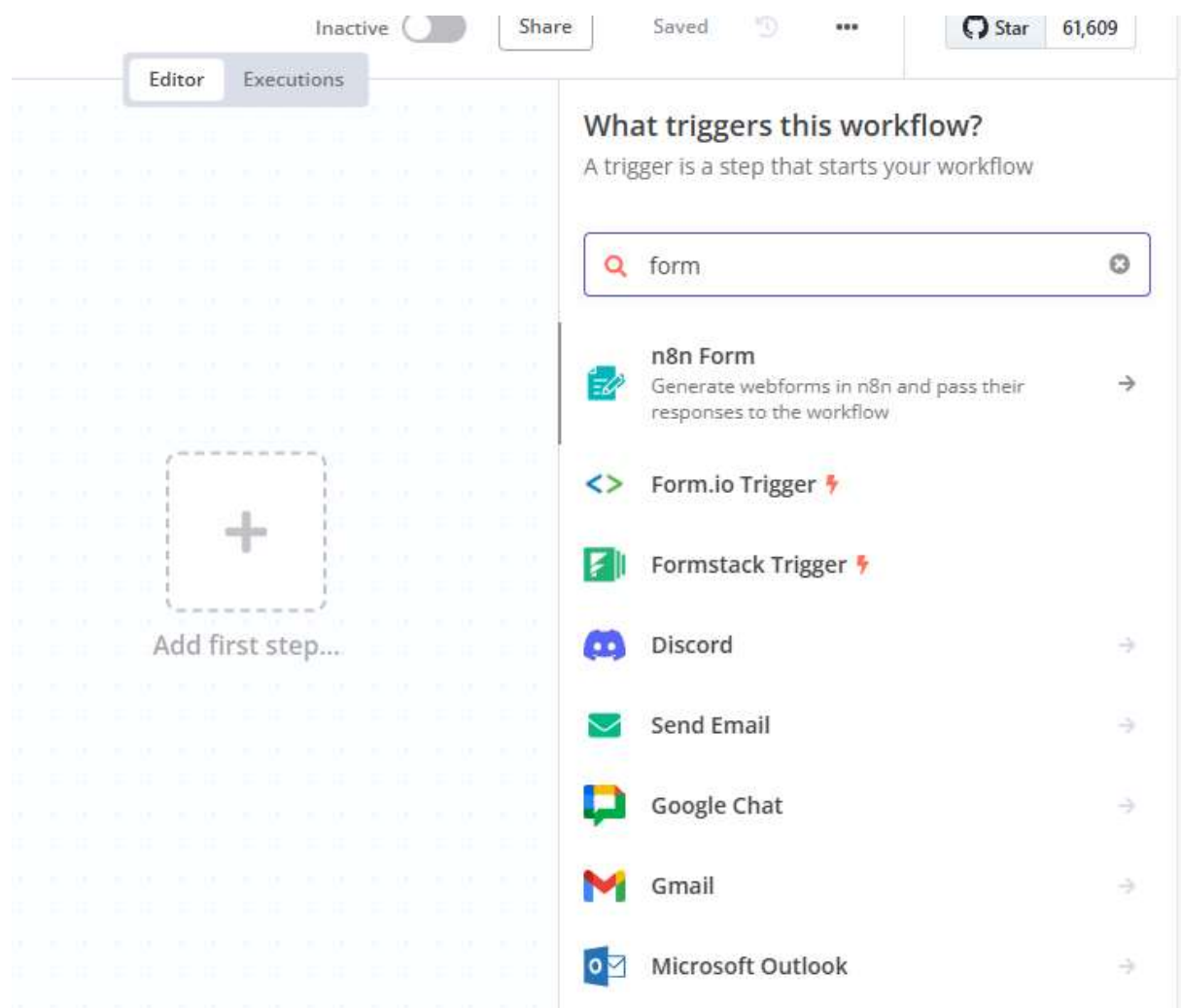
xoxb-9192442410912-9192645536416-cKpr88MQ0FNbrOhGKa8Llkpp



Figure 87: An App FORE Service is now visible

[YouTube video](#)





R. Blog writer agent

See this [video](#):

This blog writer searches a news on a topic given from the Internet using SerpAPI (Google Search) and then writes a blog. In the Blog Writer Agent, besides the System Message, also set

source for Prompt (User Message) as below. Elsewhere, I have used 'Basic LLM Chain' as a blog-writer (see AI Agent—II)

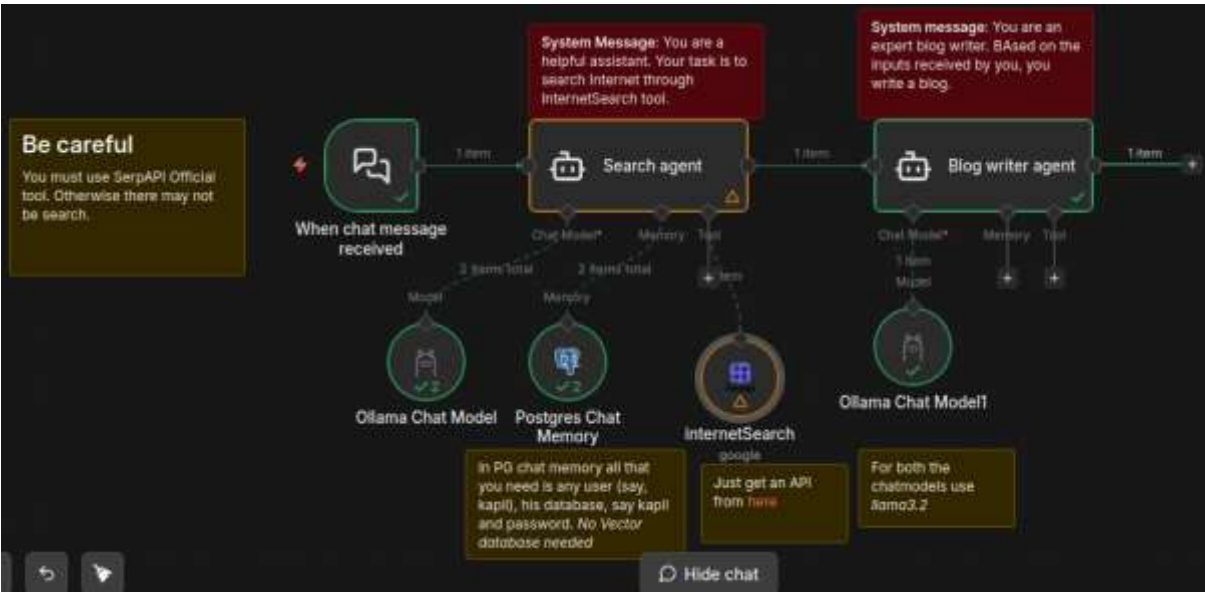


Figure 88: Blog Writer agent needs special configuration. See below.

TASK word as mentioned below should not be there.

search_metadata	search_parameters
id : 69ca4cba49579ad72c63a382 status : Success json_endpoint : https://serpapi.com/searches/v-MTR9jw5v-0-bso5kR37g/69ca4cba49579ad72c63a382.json pixel_position_endpoint : https://serpapi.com/searches/v-MTR9jw5v-0-bso5kR37g/69ca4cba49579ad72c63a382.json_with_pixel_position created_at : 2026-03-30 10:13:14 UTC processed_at : 2026-03-30 10:13:14 UTC google_url : https://www.google.com/search?q=Coffee+Write+a+blog+on+the+latest+US-Iran+war&oq=Coffee+Write+a+blog+on+the+latest+US-Iran+war&sourceid=chrome&ie=UTF-8 raw_html_file : https://serpapi.com/searches/v-MTR9jw5v-0-bso5kR37g/69ca4cba49579ad72c63a382.html total_time_taken : 4.02	engine : google q : Coffee Write a blog on the latest US-Iran war google_domain : google. device : desktop

Figure 89: Official SerpAPI tool's configuration. REMOVE 'TASK' WORD. IT DOES NOT WORK WITH IT.

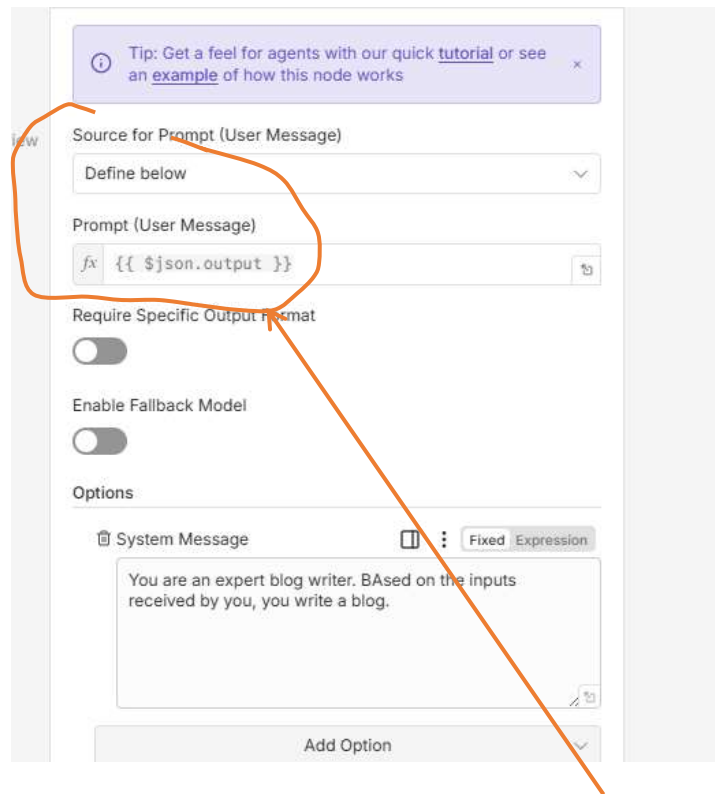


Figure 90: Blog Writer Agent: Besides System Message, make changes here.

S. HTTP Request Node

There are two *HTTP Request* nodes. One pertains to HTTP method which has an output and the other is *HTTP Request Tool* which DOES not have any output—i.e. its input comes from AI agent and output goes to AI agent.

When to use HTTP Request Tool

Below example is for Http Request node BUT not tool with an output. Use *HTTP Request Tool* when you have a URL of say a FORM or a database from where certain values are extracted which the *AI Agent* can then use for further processing..

HTTP Request Method

HTTP request node can scrap data from given web-site. But some sites, such as Wikipedia, IMDB, forbid bots. Test the site: <https://hackernoon.com/> . This site permits access. Fill up just the following in this node and click *Execute step* to see the output.

Parameters Settings

Execute step

Import cURL

Method
GET

URL
https://hackernoon.com

Authentication
None

Send Query Parameters
☐

Send Headers
☒

Specify Headers

Figure 91: Only fill up the site URL to get the site

The extracted HTML page can be fed into an HTML node, from where required anchor or tag extracted as below:

Parameters Settings

Execute step

Operation
Extract HTML Content

Source Data
JSON

JSON Property
data

Extraction Values

Key
Name

CSS Selector
h1

Return Value
Text

Deep Selections
e.g. img, div.container, #header

Return Array
☒

View

Item

- 1. The Best Medical Speech Recognition Software and APIs in 2020 (1) (The most medical speech recognition software in 2020)
- 2. The Complete Guide to Implementing Healthcare Data Agents (1) (The most complete guide to implementing healthcare data agents)
- 3. Best Speech-to-Text APIs to Build an AI Assistant in 2020 (1) (Best speech-to-text APIs to build an AI assistant in 2020)
- 4. Microsoft Print, Bing Search & other way to search - Scraping Data (1) (Microsoft Print, Bing Search & other way to search - Scraping Data)
- 5. Light-Mode
- 6. Dark-Mode

Figure 92: **key** can have any name but test CSS Selector with values as: h1, h2, h3 or body WHICHEVER gives desired result

Here is the full sequence:



Figure 93: HTTP request and HTML nodes are configured as above.

T. Webhook demo-I

Folder: C:\Users\ashok\OneDrive\Documents\n8n\13.webhook demo-II

Files: simple.html
webhook-demo-I.json

For details about webhook, read demo-II below.

The following webhook workflow is triggered by a very simple html form (code written sideways).

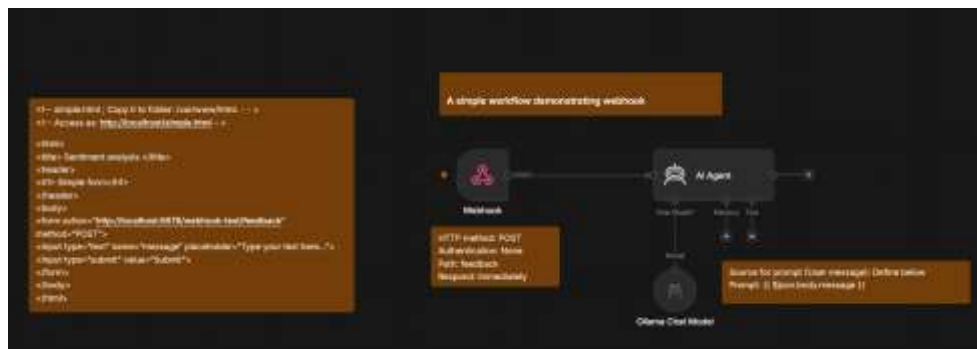


Figure 94: A simple webhook demo. IT is called by an html form on Apache webserver. Ollama chat model: llama3.2

U. Webhook demo-II

Folder: C:\Users\ashok\OneDrive\Documents\n8n\13.webhook demo-II

Files: feedback.html
Simple-feedback.html
webhook.json

A webhook essentially says: Here is my URL; call me back when an event happens and I will trigger further actions for you. A webhook is an automated, event-driven message sent between web applications. Instead of one app (App B below) constantly asking another (App A below) for updates (polling), the sending app (App A) instantly pushes data to a specific URL (endpoint) the moment a defined event occurs.

The Event: An action occurs in App A (e.g., a customer makes a payment or submits a form).

The Payload: App A gathers the relevant data about that event (usually in JSON format).

The Delivery: App A fires an HTTP POST request, sending that payload directly to a specific URL provided by App B. App B has a webhook.

In the context of webhooks, a **callback** (specifically an "HTTP callback") is the exact **URL or destination address** that you provide to the sending application (App A) so it knows where to deliver the data.

When you configure a webhook, you are essentially telling the App A application: "When this event happens, call me (i.e. App B) back at this specific web address."



Figure 95: The flow. Call Ollama is HTTP Request node

Here are the configurations:

My prompt on Claude Desktop:

I am working on my Intranet. I have n8n installed on my laptop. I also have ollama installed as also Apache webserver and PostgreSQL installed. I want to design a simple n8n experiment to demonstrate the webhook usage. I do not have api access to Gmail or google sheets or to other google products.

Response:

This is a great use of what you already have — you can build a fully local, self-contained webhook demo without touching any Google service. Here's a clean design that exercises all four tools together: Apache serves a simple form, n8n's Webhook node receives the submission, Ollama analyses the text, and PostgreSQL stores the result.

The idea

An HTML feedback form (served by Apache) posts text to an **n8n Webhook**. n8n forwards that text to **Ollama** for analysis (e.g. sentiment or a one-line summary), writes the original text plus Ollama's response into **PostgreSQL**, and then replies to the original request with a JSON result. Every piece of software you already have gets used, and nothing leaves your laptop.

1. Postgres: create a table

```
CREATE TABLE feedback_log (  
    id SERIAL PRIMARY KEY,  
    submitted_text TEXT NOT NULL,  
    ollama_response TEXT,  
    created_at TIMESTAMP DEFAULT NOW()  
);
```

Proceed as follow to create table in database 'harnal', user: Harnal, password: harnal:



```
ashok@ashok:~$ sudo su harnal  
[sudo] password for ashok:  
harnal@ashok:~$ psql  
could not change directory to "/home/ashok": Permission denied  
psql (14.23 (Ubuntu 14.23-0ubuntu0.22.04.1))  
Type "help" for help.  
  
harnal=> \c harnal  
You are now connected to database "harnal" as user "harnal".  
harnal=>
```

Figure 96: Create table now...

2. n8n workflow

Node 1 — Webhook

- HTTP Method: POST
 - Path: feedback
 - Response Mode: Using 'Respond to Webhook' Node
- This gives you a production URL like <http://localhost:5678/webhook/feedback> OR a test URL like: <http://localhost:5678/webhook-test/feedback>

Node 2 — HTTP Request (call Ollama)

- Method: POST
- URL: <http://localhost:11434/api/generate>
- Body (JSON):

```
{  
    "model": "llama3.2",  
    "prompt": "In one short sentence, describe the sentiment of this feedback: {{ $json.body.text }}",  
    "stream": false  
}
```

(swap llama3.2 for whatever model you've pulled in Ollama — check with `ollama list`)

Node 3 — Postgres (Insert)

- Operation: Insert
- Table: feedback_log
- Columns:
 - submitted_text → {{ \$('webhook').item.json.body.text }}
 - ollama_response → {{ \$json.response }}

Node 4 — Respond to Webhook

- Response Body:


```
{
    "status": "ok",
    "analysis": "{{ $('HTTP Request').item.json.response }}"
  }
```

3. Apache: the form

A tiny static page (feedback.html) served by Apache that posts to the n8n webhook with fetch:

```
$ sudo cp feedback.html /var/www/html/

<!DOCTYPE html>
<html>
<body>
  <h3>Leave feedback</h3>
  <textarea id="text" rows="4" cols="40"></textarea><br>
  <button onclick="submitFeedback()">Submit</button>
  <p id="result"></p>

  <script>
    async function submitFeedback() {
      const text = document.getElementById('text').value;
      const res = await fetch('http://localhost:5678/webhook/feedback', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ text })
      });
      const data = await res.json();
      document.getElementById('result').innerText = data.analysis;
    }
  </script>
</body>
</html>
```

Drop this in Apache's document root (e.g. /var/www/html/feedback.html), and it's live on your intranet and access as: <http://localhost/feedback.html>.

If you are having *Test-URL* and not *production* URL for webhook, modify one line within the script as:

```
const res = await fetch('http://localhost:5678/webhook-test/feedback',
```

4. Test without the form first

Before wiring up Apache, just hit the webhook directly write in Ubuntu terminal and press ENTER to confirm that the n8n side works:

```
curl -X POST http://localhost:5678/webhook/feedback \
-H "Content-Type: application/json" \
-d '{"text": "The new intranet page is really slow to load."}'
```

You should get back JSON with Ollama's take on the sentiment, and a new row in feedback_log.

4. What does the function inside the script do?

This function runs when someone clicks "Submit" on the feedback form, and it walks through the whole round trip to your n8n workflow. Step by step:

1. **const text = document.getElementById('text').value;** Grabs whatever the user typed into the <textarea id="text"> on the page.
2. **const res = await fetch('http://localhost:5678/webhook/feedback', {...})** Sends that text as an HTTP POST request to your n8n webhook. The await pauses this function until n8n finishes processing and sends back a response — so it waits through the Ollama call and the Postgres insert before continuing.
 - method: 'POST' — this is a POST request, matching the Webhook node's configured method.
 - headers: { 'Content-Type': 'application/json' } — tells n8n the body is JSON, not plain text or a form.
 - body: JSON.stringify({ text }) — packages the text into a JSON object like {"text": "the page is slow"} and converts it to a string, since fetch bodies have to be strings, not raw objects.
3. **const data = await res.json();** Takes the response n8n sends back (from your Respond to Webhook node, which returns something like {"status": "ok", "analysis": "..."}) and parses it from a JSON string into a JavaScript object.
4. **document.getElementById('result').innerText = data.analysis;** Pulls the analysis field out of that object — Ollama's sentiment description — and writes it into the <p id="result"> element on the page, so the user sees the result without the page reloading.

One gap worth flagging: there's no error handling. If the webhook is down, returns a non-200 status, or data.analysis is undefined for some reason, this will either throw an unhandled error or silently write "undefined" into the result box..

If you're fine with the page navigating (or reloading) after submit, you don't need JS or fetch:

```
$ sudo cp simple-feedback.html /var/www/html/

<form action="http://localhost:5678/webhook/feedback" method="POST">
  <textarea name="text" rows="4" cols="40"></textarea>
  <button type="submit">Submit</button>
</form>
```

The browser handles the POST itself. The catch: it'll navigate to the webhook URL and show n8n's raw JSON response as the new page, unless you set the form's target to a hidden <iframe> to suppress that.

Flow architecture

The full architecture. Note the loop of response.

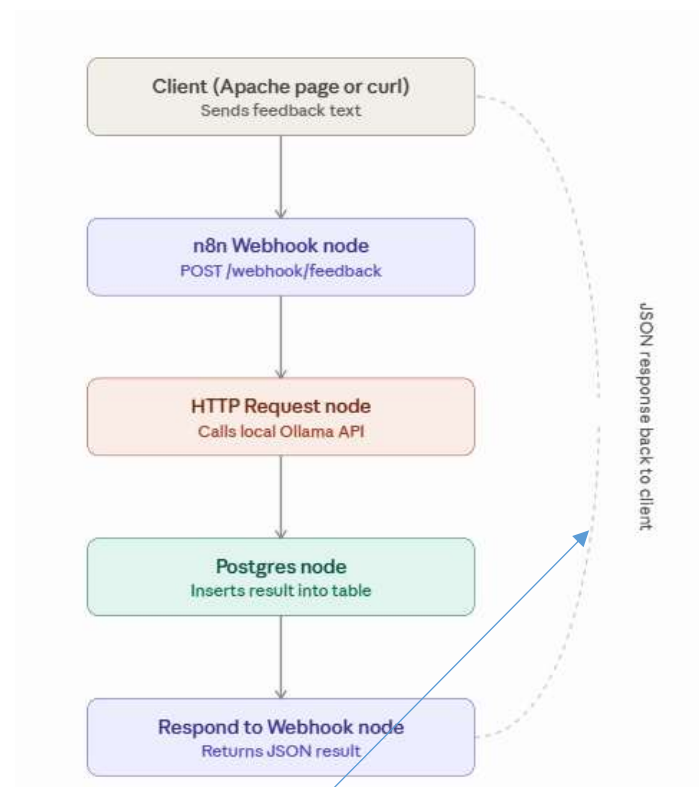


Figure 97: Note the response back to client

The path: you typed *feedback* into the Webhook node, which is why the URL ends in */webhook/feedback*. If you'd typed *submit-form* instead, that would be the URL.

Parameters Settings Listen for test event

Webhook URL

Test URL Production URL

http://localhost:5678/webhook-test/feedback

HTTP Method: POST

Path: feedback

Authentication: None

Respond: Using 'Respond to Webhook' Node

Insert a 'Respond to Webhook' node to control when and how you respond. [Show details](#)

Options: No properties

Add option

Figure 98: webhook node configuration.

Note: it is 'webhook-test/feedback' AND not 'webhook/feedback'

Method: POST

URL: http://localhost:11434/api/generate

Authentication: None

Send Query Parameters: ☐

Send Headers: ☐

Send Body: ☒

Body Content Type: JSON

Specify Body: Using JSON

JSON

```
{
  "model": "llama3.2",
  "prompt": "In one short sentence, describe the sentiment of this feedback: {{ $body.text }}",
  "stream": false
}
```

Options: No properties

Figure 99: HTTP Request node calling ollama URL



Figure 100: Postgres node



Figure 101: Respond to webhook node

In n8n, a node **does not automatically receive the outputs of all previous nodes** in the workflow. Instead, a node only receives the input from the immediately preceding node to which it is connected. If you need data from a node that is not immediately before the current node, you can use **expressions** to reference it specifically by its node name (as above):

- **Syntax:** Use `{{ ${{ 'Node Name' }}.item.json }}` to pull data from a specific previous step.
- **All Data:** Use `{{ ${{ 'Node Name' }}.all() }}` to access all items from that node.

V. Webhook triggered by postgres

Folder: D:\OneDrive\Documents\n8n\16.webhook_postgres_demo

A webhook in n8n can get triggered when a row of data is inserted into a table. Extensive write is given in our file: *'postgres trigger for webhook in n8n.txt'*. Also see my conversation with Google AI in file: *'postgresql data insertion trigger.pdf'*.

W. DataTable

Also refer [this link](#)

Nodes used (in sequence):

- Form node: Collects: Name, email, Question, Answer
- If node: Checks if email is from domain: *fsm.ac.in*
- Edit node or Set node: Appends a field: *isTrusted* to input data received from the form. If email contains domain *fsm.ac.in*, value of *isTrusted* is true else it is false.
- No Op node: It merely forwards its input to output. *No Op* node has two inputs. Out of the two inputs, only one of them would be having some data. So, the *No Op* node forwards that input only.
- Basic LLM Chain: It considers two inputs: Question and Answer and generates relevant tags.

- f. DataTable → A Data table must pre-exist containing five columns by ANY name: Name, email, Question, Answer and tags. Form data fields, *isTrusted* field and tags get inserted as one row.



Figure 102: *isTrusted* node on true branch of if, appends a column *isTrusted* with value true. On the false branch value added is false.

Our DataTable name is *Data*. The DataTable panel is as below:

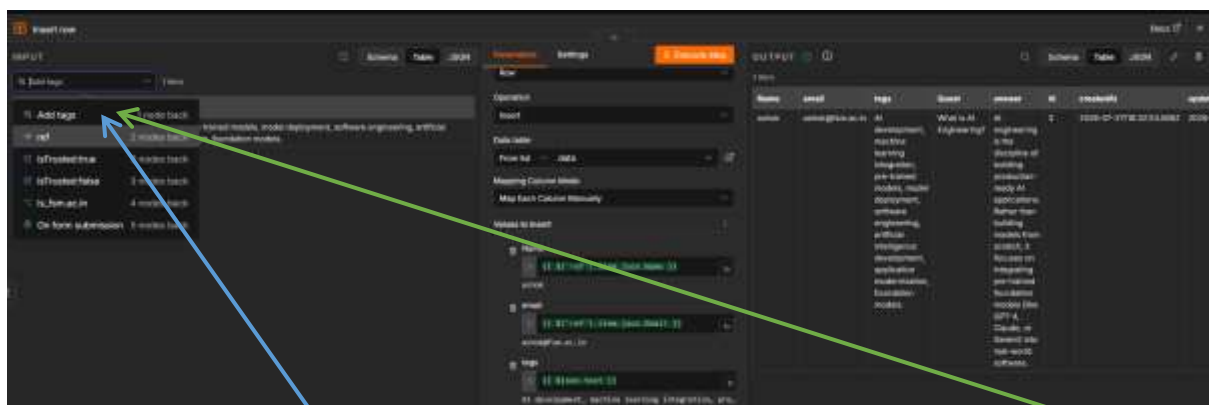


Figure 103: Columns in DataTable can receive inputs not only from the Basic LLM Chain but also from anyone of the previous nodes. (See below the expanded view).

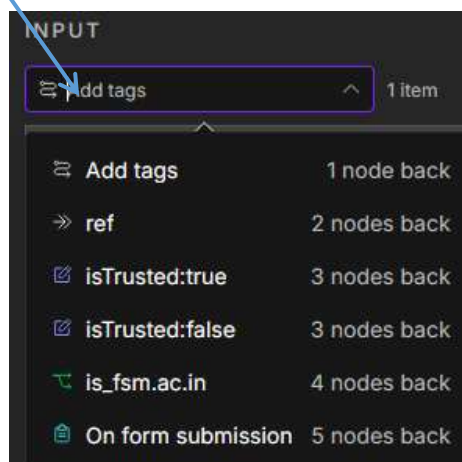


Figure 104: Input to DataTable columns can be from any one of the above nodes.

See DataTable configuration below.

Operation

Insert

Data table

From list data

Mapping Column Mode

Map Each Column Manually

Values to Insert

Name

fx

{{ \$('ref').item.json.Name }}

ashok

email

fx

{{ \$('ref').item.json.Email }}

ashok@fsm.ac.in

tags

fx

{{ \$json.text }}

AI development, machine learning integration, pre...

Quest

fx

{{ \$('ref').item.json.Question }}

What is AI Engineering?

answer

fx

{{ \$('ref').item.json.Answer }}

AI engineering is the discipline of building prod...

Options

Figure 105: DataTable name is Data. Note that one column name in DataTable is 'Quest' but form has output label as 'Question'.